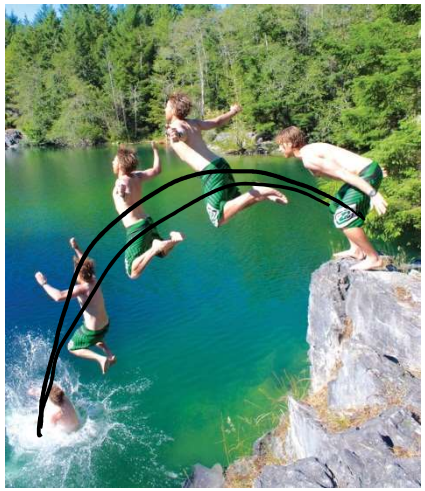
Storyline

Chapter 4:
Motion in Two Dimensions







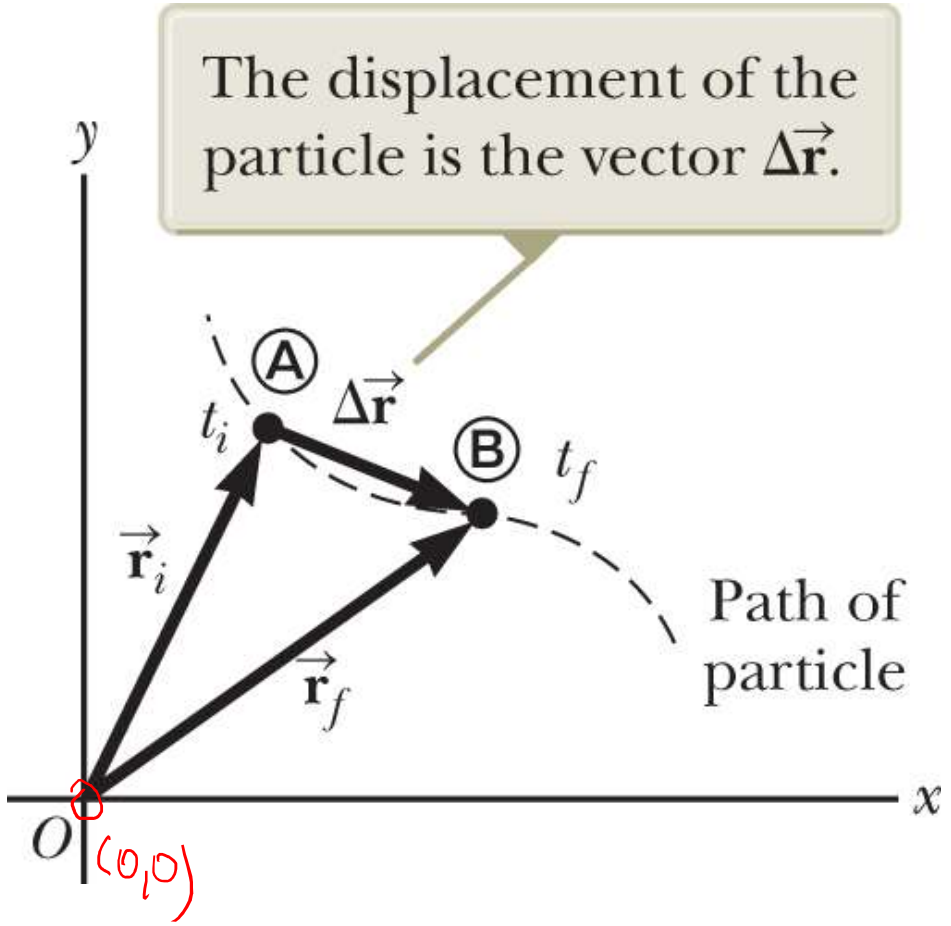


Physics for Scientists and Engineers, 10e
Raymond A. Serway
John W. Jewett, Jr.

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Position Vector

$$\Delta\vec{\mathbf{r}} \equiv \vec{\mathbf{r}}_f - \vec{\mathbf{r}}_i$$



Velocity Vector

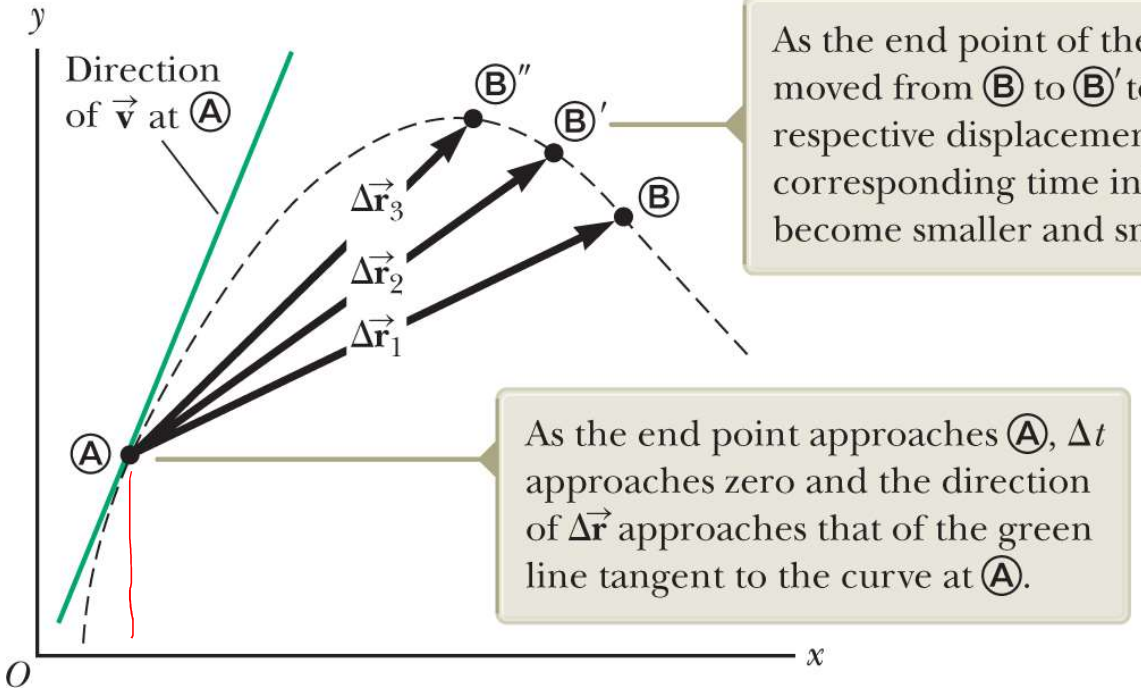
Instantaneous velocity

Average velocity

$$\vec{\mathbf{v}}_{\text{avg}} \equiv \frac{\Delta \vec{\mathbf{r}}}{\Delta t}$$

$$\vec{\mathbf{v}} \equiv \lim_{\Delta t \to 0} \frac{\Delta \vec{\mathbf{r}}}{\Delta t} = \frac{d\vec{\mathbf{r}}}{dt}$$

$$v = |\vec{\mathbf{v}}|$$



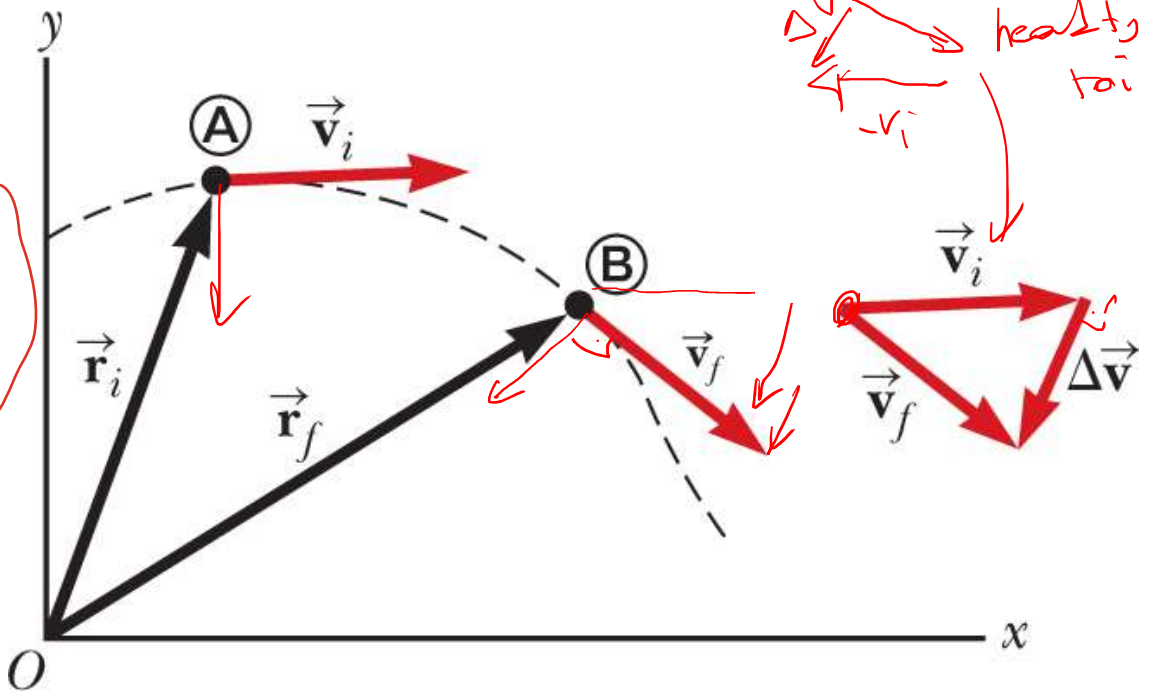
As the end point of the path is moved from Ⓑ to Ⓑ′ to Ⓑ″, the respective displacements and corresponding time intervals become smaller and smaller.

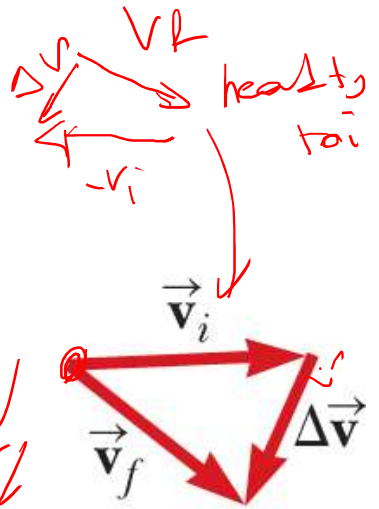
As the end point approaches Ⓐ, Δt approaches zero and the direction of $\Delta\vec{\mathbf{r}}$ approaches that of the green line tangent to the curve at Ⓐ.

Acceleration Vector

$$\vec{\mathbf{a}}_{\text{avg}} \equiv \frac{\Delta\vec{\mathbf{v}}}{\Delta t} = \frac{\vec{\mathbf{v}}_f - \vec{\mathbf{v}}_i}{t_f - t_i}$$

$$\vec{\mathbf{a}} \equiv \lim_{\Delta t \to 0} \frac{\Delta\vec{\mathbf{v}}}{\Delta t} = \frac{d\vec{\mathbf{v}}}{dt}$$





Quick Quiz 4.1

Consider the following controls in an automobile in motion: gas pedal, brake, steering wheel. What are the controls in this list that cause an acceleration of the car?

(a) all three controls

(b) the gas pedal and the brake

(c) only the brake

(d) only the gas pedal

(e) only the steering wheel

Quick Quiz 4.1

Consider the following controls in an automobile in motion: gas pedal, brake, steering wheel. What are the controls in this list that cause an acceleration of the car?

(a) all three controls

(b) the gas pedal and the brake

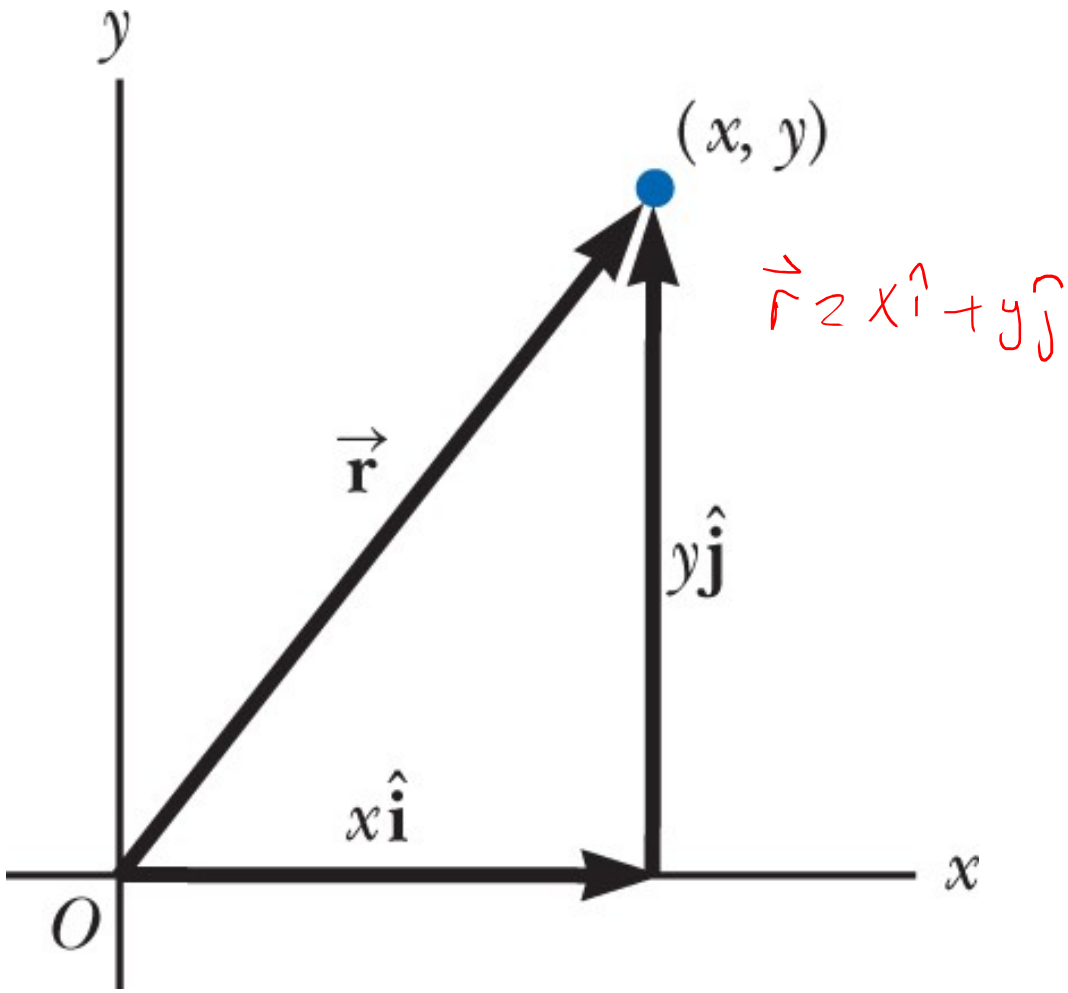
(c) only the brake

(d) only the gas pedal

(e) only the steering wheel

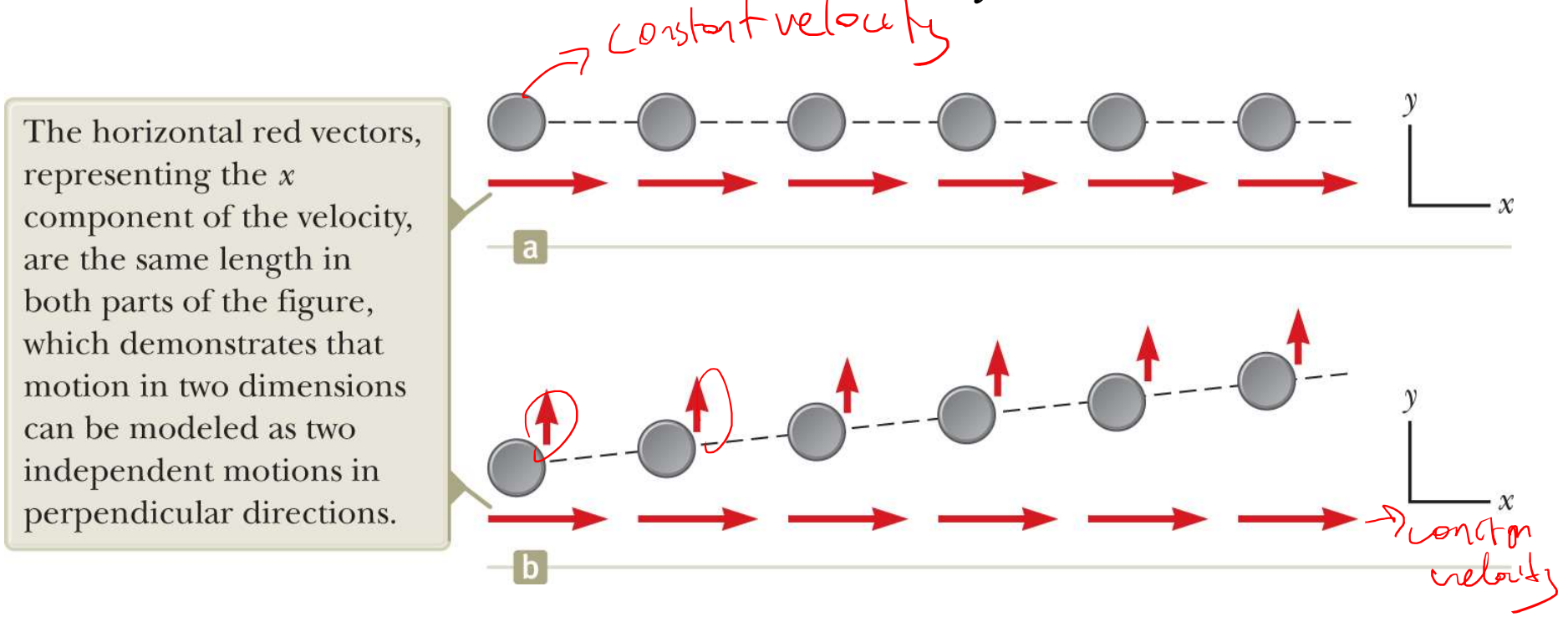
Two-Dimensional Motion with Constant Acceleration

$$\vec{\mathbf{r}} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}}$$



Two-Dimensional Motion with Constant Acceleration

Motion in two dimensions can be modeled as two *independent* motions in each of the two perpendicular directions associated with the x and y axes.



The horizontal red vectors, representing the x component of the velocity, are the same length in both parts of the figure, which demonstrates that motion in two dimensions can be modeled as two independent motions in perpendicular directions.

Two-Dimensional Motion with Constant Acceleration

$$\vec{\mathbf{v}} = \frac{d\vec{\mathbf{r}}}{dt} = \frac{dx}{dt}\hat{\mathbf{i}} + \frac{dy}{dt}\hat{\mathbf{j}} = v_x\hat{\mathbf{i}} + v_y\hat{\mathbf{j}}$$

$$v_{xf} = v_{xi} + a_x t \quad \text{and} \quad v_{yf} = v_{yi} + a_y t$$

$$\vec{\mathbf{v}}_f = \left(v_{xi} + a_x t\right)\hat{\mathbf{i}} + \left(v_{yi} + a_y t\right)\hat{\mathbf{j}} = \left(v_{xi}\hat{\mathbf{i}} + v_{yi}\hat{\mathbf{j}}\right) + \left(a_x\hat{\mathbf{i}} + a_y\hat{\mathbf{j}}\right)t$$

$$\vec{\mathbf{v}}_f = \vec{\mathbf{v}}_i + \vec{\mathbf{a}}t \quad \left(\text{for constant } \vec{\mathbf{a}}\right)$$

Two-Dimensional Motion with Constant Acceleration

$$x_f = x_i + v_{xi}t + \frac{1}{2}a_x t^2 \qquad y_f = y_i + v_{yi}t + \frac{1}{2}a_y t^2$$

$$\vec{\mathbf{r}}_f = \left(x_i + v_{xi}t + \frac{1}{2}a_x t^2\right)\hat{\mathbf{i}} + \left(y_i + v_{yi}t + \frac{1}{2}a_y t^2\right)\hat{\mathbf{j}}$$

$$= \left(x_i\hat{\mathbf{i}} + y_i\hat{\mathbf{j}}\right) + \left(v_{xi}\hat{\mathbf{i}} + v_{yi}\hat{\mathbf{j}}\right)t + \frac{1}{2}\left(a_x\hat{\mathbf{i}} + a_y\hat{\mathbf{j}}\right)t^2$$

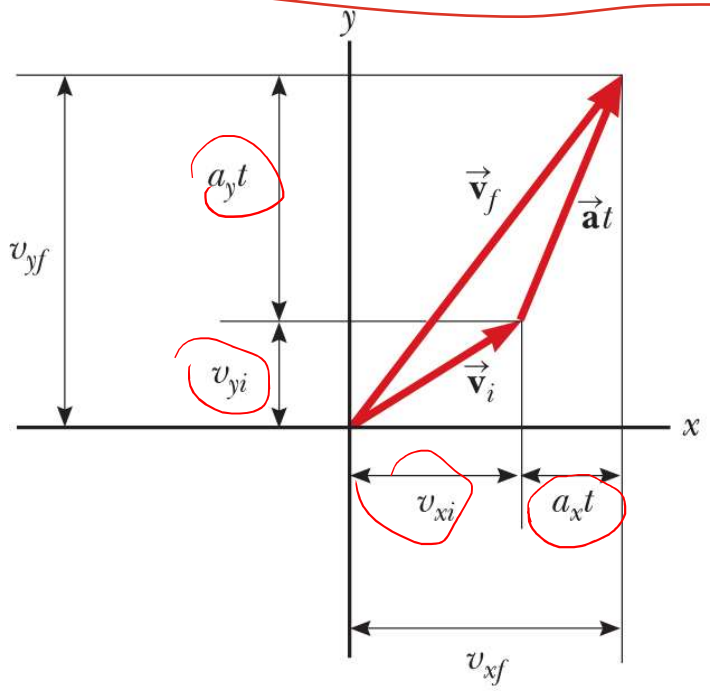
$$\vec{\mathbf{r}}_f = \vec{\mathbf{r}}_i + \vec{\mathbf{v}}_i t + \frac{1}{2}\vec{\mathbf{a}}t^2 \quad \left(\text{for constant } \vec{\mathbf{a}}\right)$$

Two-Dimensional Motion with Constant Acceleration

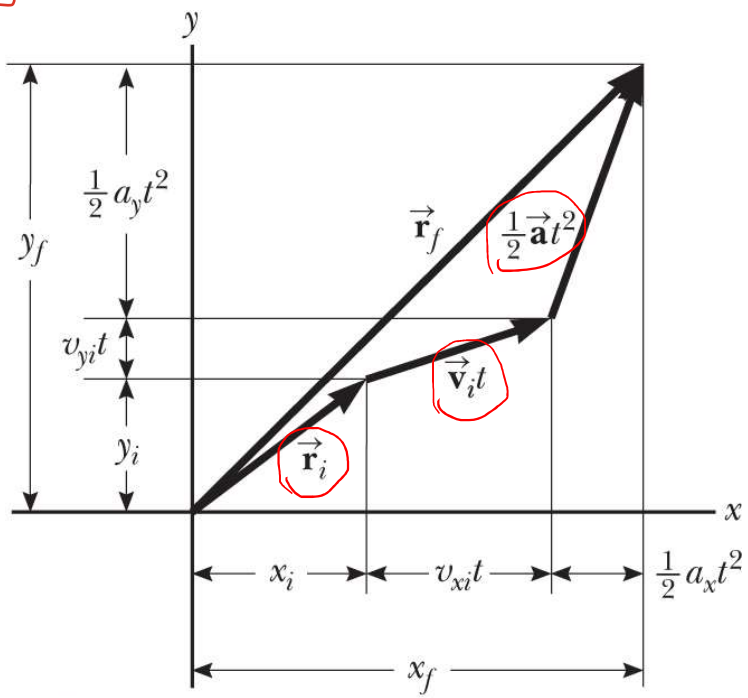
$$\vec{\mathbf{v}}_f = \vec{\mathbf{v}}_i + \vec{\mathbf{a}}t$$

$$\vec{\mathbf{r}}_f = \vec{\mathbf{r}}_i + \vec{\mathbf{v}}_i t + \frac{1}{2}\vec{\mathbf{a}}t^2$$

(for constant $\vec{\mathbf{a}}$)



a



b

Example 4.1: Motion in a Plane

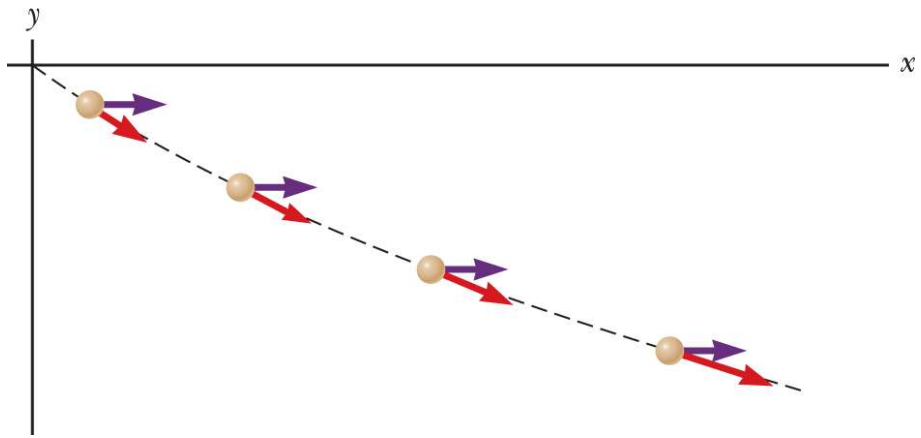
A particle moves in the *xy* plane, starting from the origin at $t = 0$ with an initial velocity having an *x* component of 20 m/s and a *y* component of –15 m/s. The particle experiences an acceleration in the *x* direction, given by $a_x = 4.0\ \text{m/s}^2$.

$v_{xi} = 20\ \text{m/s}$

$v_{yi} = -15\ \text{m/s}$

$a_x = 4\ \text{m/s}^2$

(A) Determine the total velocity vector at any later time.



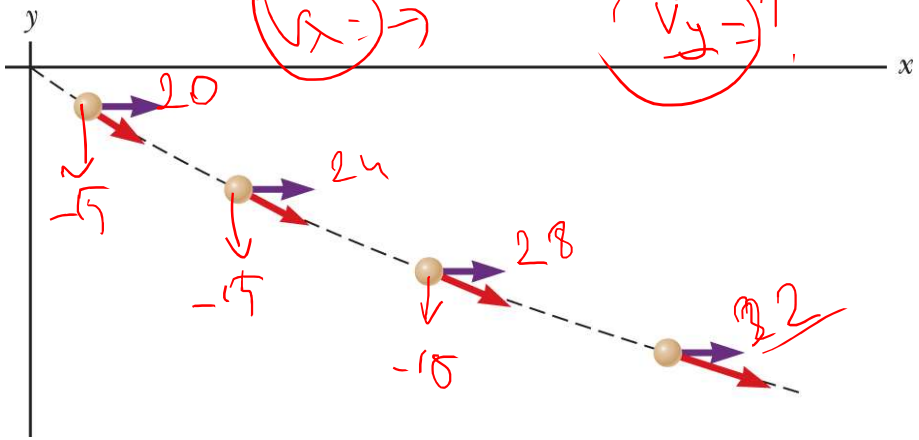
Example 4.1: Motion in a Plane

$$v_{xi} = 20 \text{ m/s}, \; v_{yi} = -15 \text{ m/s}, \; a_x = 4.0 \text{ m/s}, \; a_y = 0$$

$$\vec{\mathbf{v}}_f = \vec{\mathbf{v}}_i + \vec{\mathbf{a}}t = \left(v_{xi} + a_x t\right)\hat{\mathbf{i}} + \left(v_{yi} + a_y t\right)\hat{\mathbf{j}}$$

$$\vec{\mathbf{v}}_f = \left[20 + (4.0)t\right]\hat{\mathbf{i}} + \left[-15 + (0)t\right]\hat{\mathbf{j}}$$

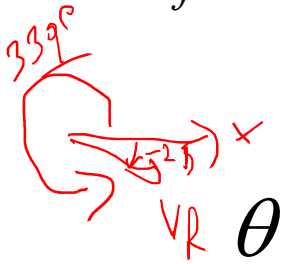
$$\vec{\mathbf{v}}_f = \boxed{\left[20 + (4.0)t\right]\hat{\mathbf{i}} - 15\hat{\mathbf{j}}}$$



Example 4.1: Motion in a Plane

(B) Calculate the velocity and speed of the particle at $t = 5.0$ s and the angle the velocity vector makes with the x axis.

$$\vec{\mathbf{v}}_f = \left[\left(20 + 4.0(5.0)\right)\hat{\mathbf{i}} - 15\hat{\mathbf{j}}\right] = \boxed{\left(40\hat{\mathbf{i}} - 15\hat{\mathbf{j}}\right)\ \text{m/s}}$$



$$\theta = \tan^{-1}\left(\frac{v_{yf}}{v_{xf}}\right) = \tan^{-1}\left(\frac{-15\ \text{m/s}}{40\ \text{m/s}}\right) = -21°$$

$$v_f = \left|\vec{\mathbf{v}}_f\right| = \sqrt{{v_{xf}}^2 + {v_{yf}}^2} = \sqrt{(40)^2 + (-15)^2} = \boxed{43\ \text{m/s}}$$

Example 4.1: Motion in a Plane

(C) Determine the x and y coordinates of the particle at any time t and its position vector at this time.

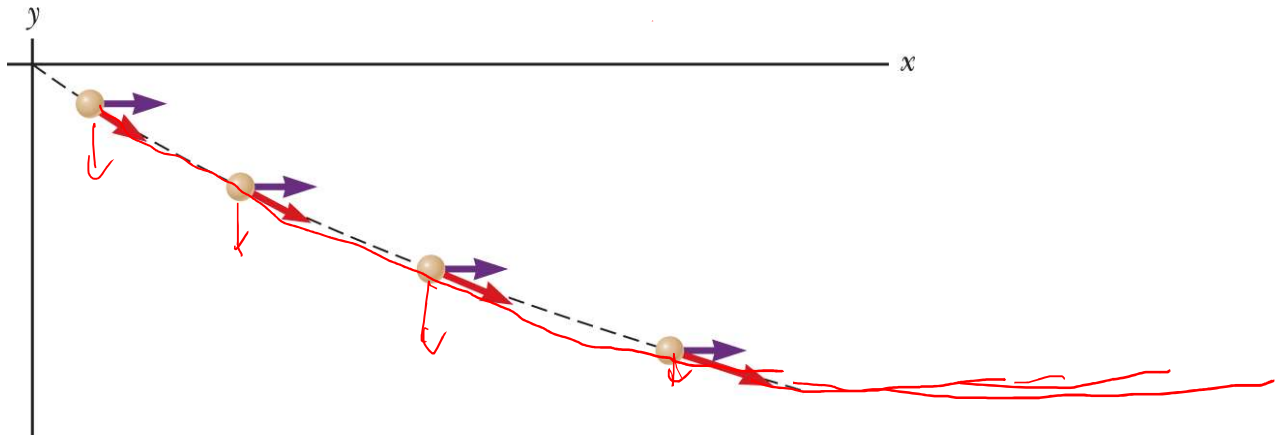
$$\vec{\mathbf{r}}_f = \vec{\mathbf{r}}_i + \vec{\mathbf{v}}_i t + \frac{1}{2}\vec{\mathbf{a}}t^2 = \left(r_{xi} + v_{xi}t + \frac{1}{2}a_x t^2\right)\hat{\mathbf{i}} + \left(r_{yi} + v_{yi}t + \frac{1}{2}a_y t^2\right)\hat{\mathbf{j}}$$

$$\vec{\mathbf{r}}_f = \left[0 + (20)t + \frac{1}{2}(4.0)t^2\right]\hat{\mathbf{i}} + \left[0 + (-15)t + \frac{1}{2}(0)t^2\right]\hat{\mathbf{j}}$$

$$\boxed{\vec{\mathbf{r}}_f = \left(20t + 2.0t^2\right)\hat{\mathbf{i}} - 15t\hat{\mathbf{j}}}$$

Example 4.1:
Motion in a Plane

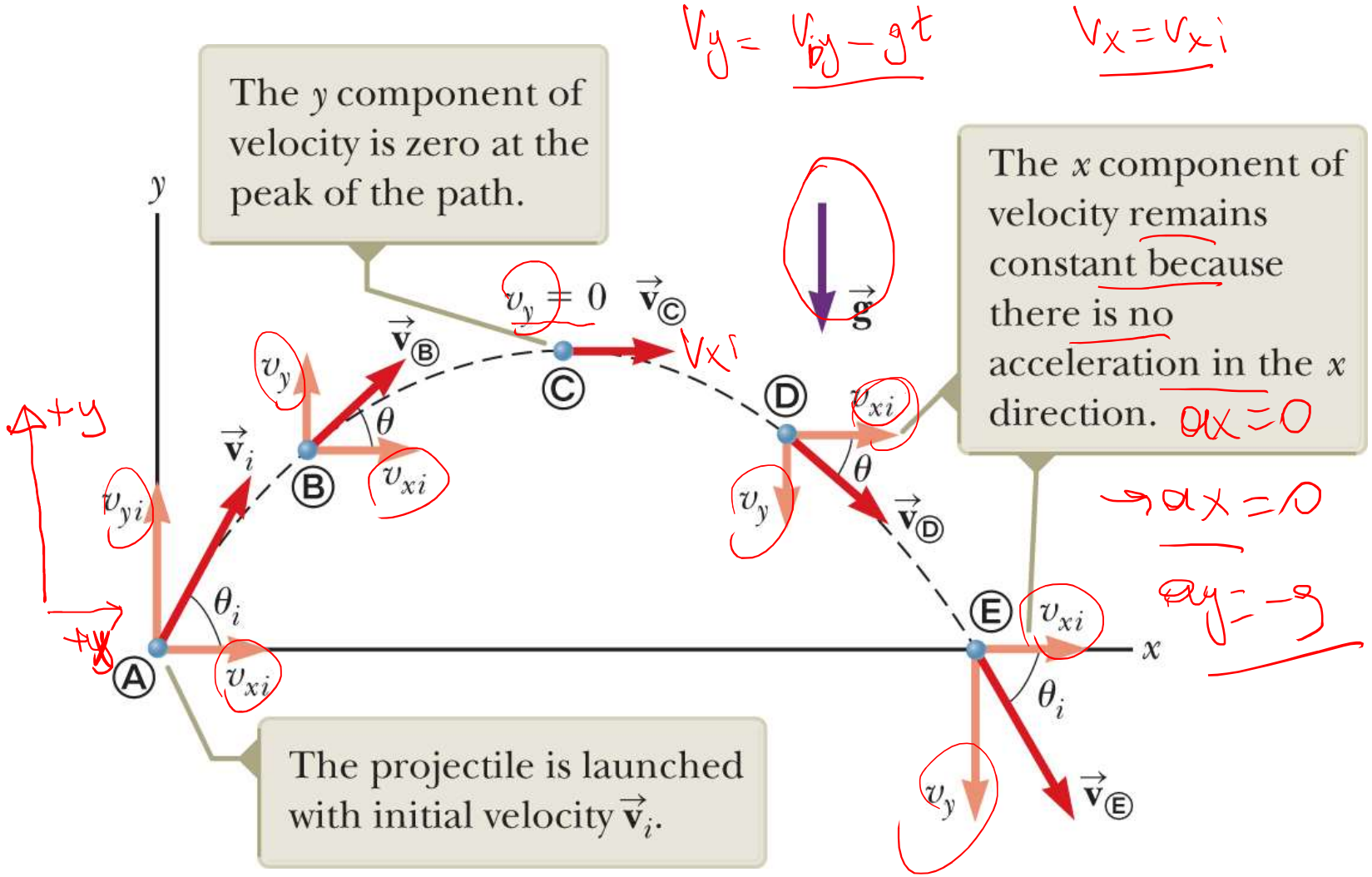
What if we wait a very long time and then observe the motion of the particle? How would we describe the motion of the particle for large values of the time?



Path becomes more and more parallel to *x* axis.

$$\vec{\mathbf{v}}_f = \left[20 + (4.0)t\right]\hat{\mathbf{i}} - 15\hat{\mathbf{j}}$$

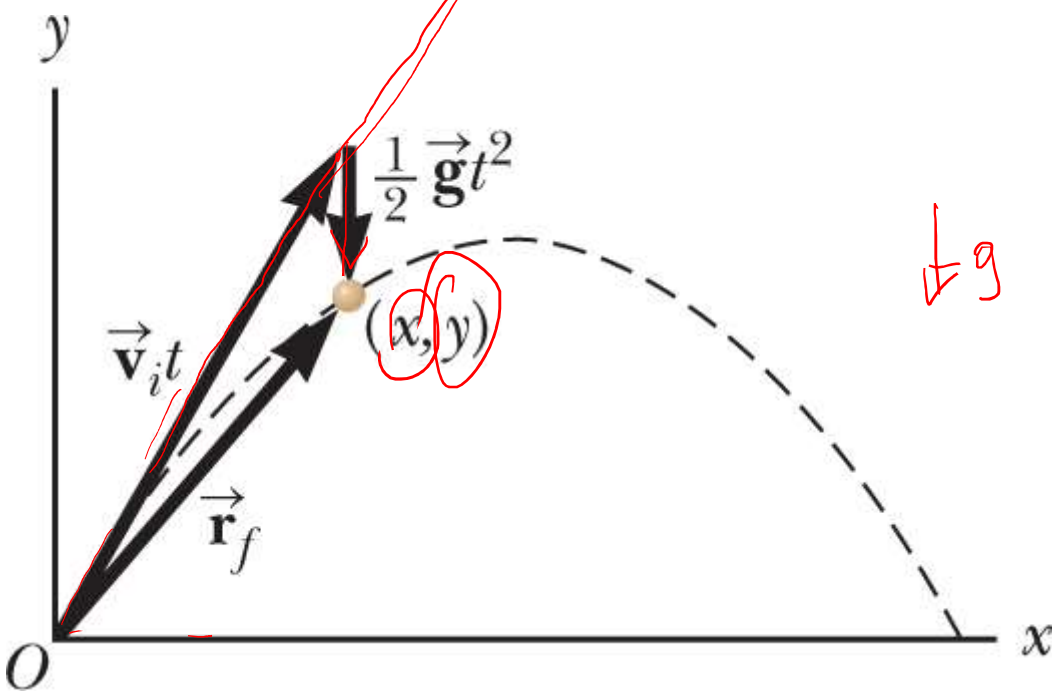
Projectile Motion



Projectile Motion

$$\vec{\mathbf{r}}_f = \vec{\mathbf{r}}_i + \vec{\mathbf{v}}_i t + \frac{1}{2}\vec{\mathbf{g}}t^2$$

$$v_{xi} = v_i \cos\theta_i \qquad v_{yi} = v_i \sin\theta_i$$

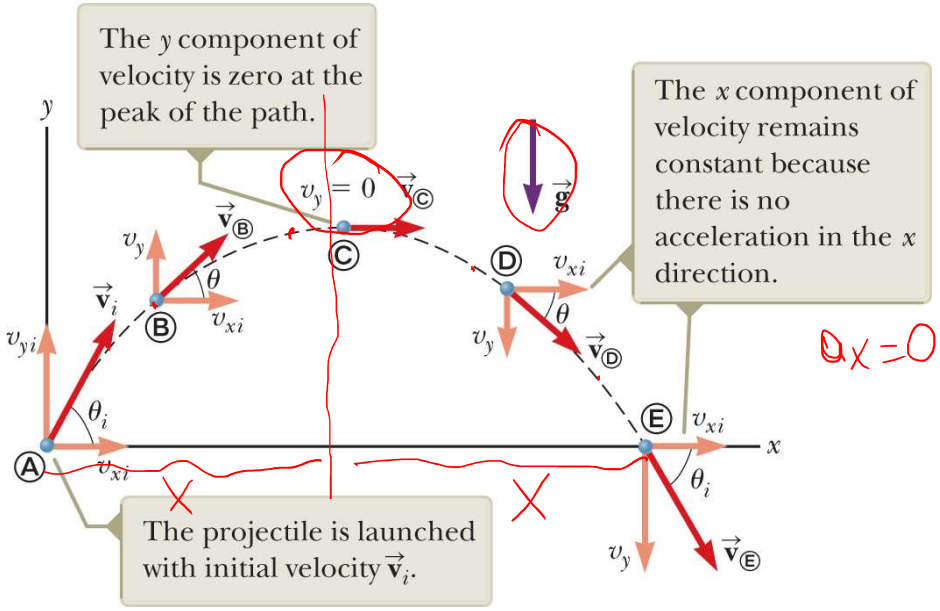


Projectile Motion

$$x_f = x_i + v_{xi}t$$

$$v_{yf} = v_{yi} - gt$$

$$v_{y,avg} = \frac{v_{yi} + v_{yf}}{2}$$



$$y_f = y_i + \frac{1}{2}\left(v_{yi} + v_{yf}\right)t$$

$$y_f = y_i + v_{yi}t - \frac{1}{2}gt^2$$

$$v_{yf}^{\,2} = v_{yi}^{\,2} - 2g\left(y_f - y_i\right)$$

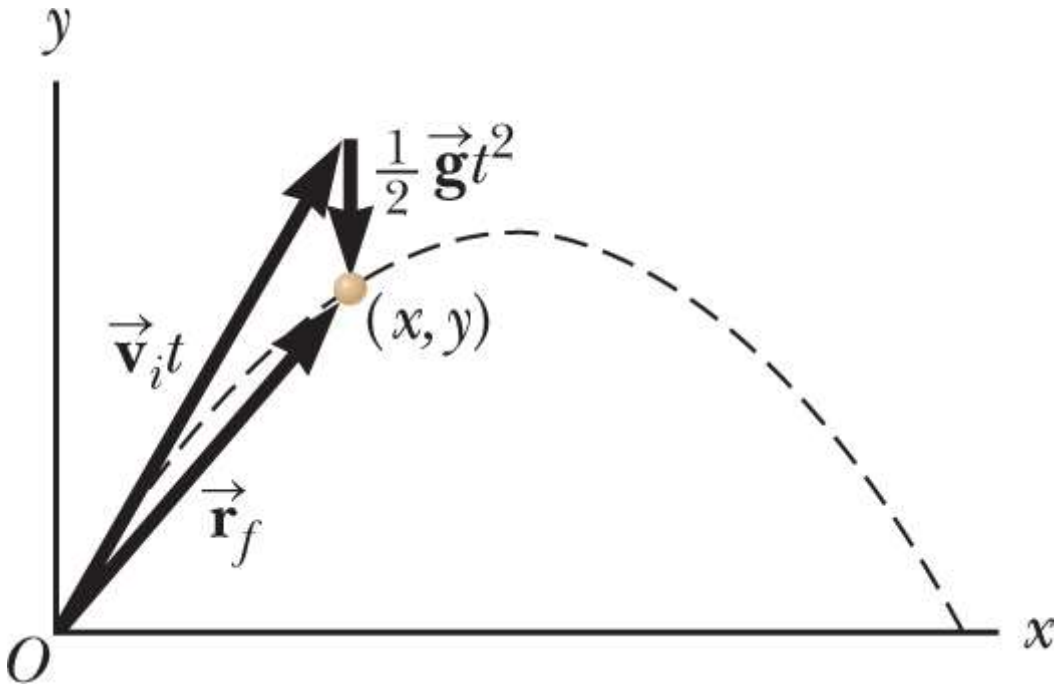
Quick Quiz 4.2 Part I

As a projectile thrown at an upward angle moves in its parabolic path (such as in the figure), at what point along its path are the velocity and acceleration vectors for the projectile perpendicular to each other?

(a) nowhere

(b) the highest point

(c) the launch point



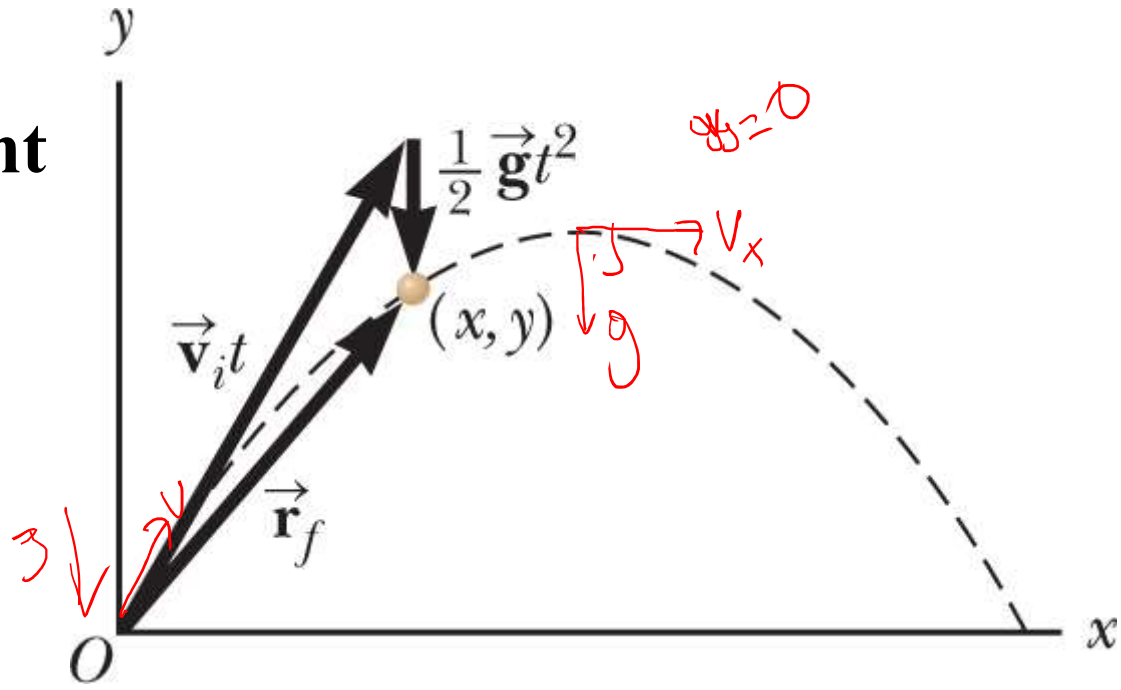
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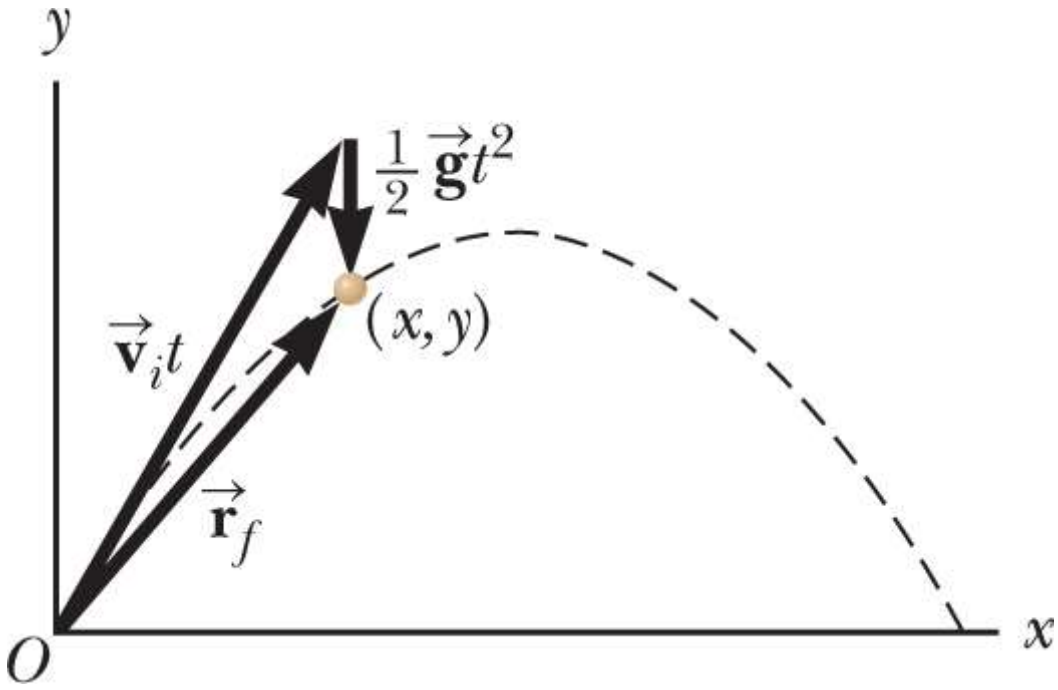
Quick Quiz 4.2 Part II

At what point are the velocity and acceleration vectors for the projectile parallel to each other?

(a) nowhere

(b) the highest point

(c) the launch point



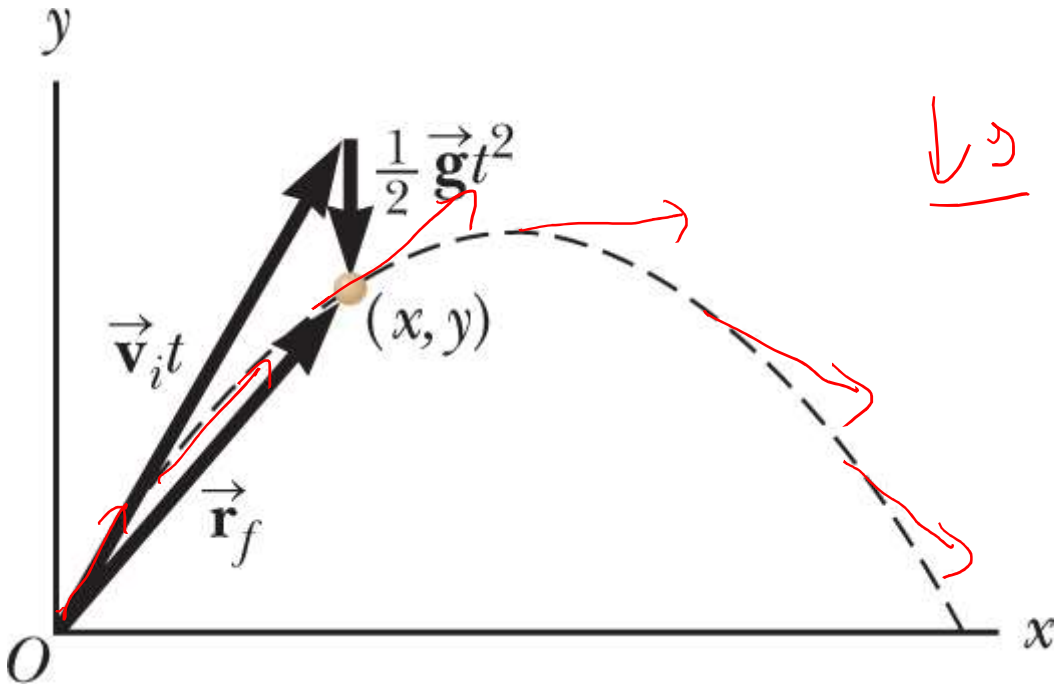
Quick Quiz 4.2 Part II

At what point are the velocity and acceleration vectors for the projectile parallel to each other?

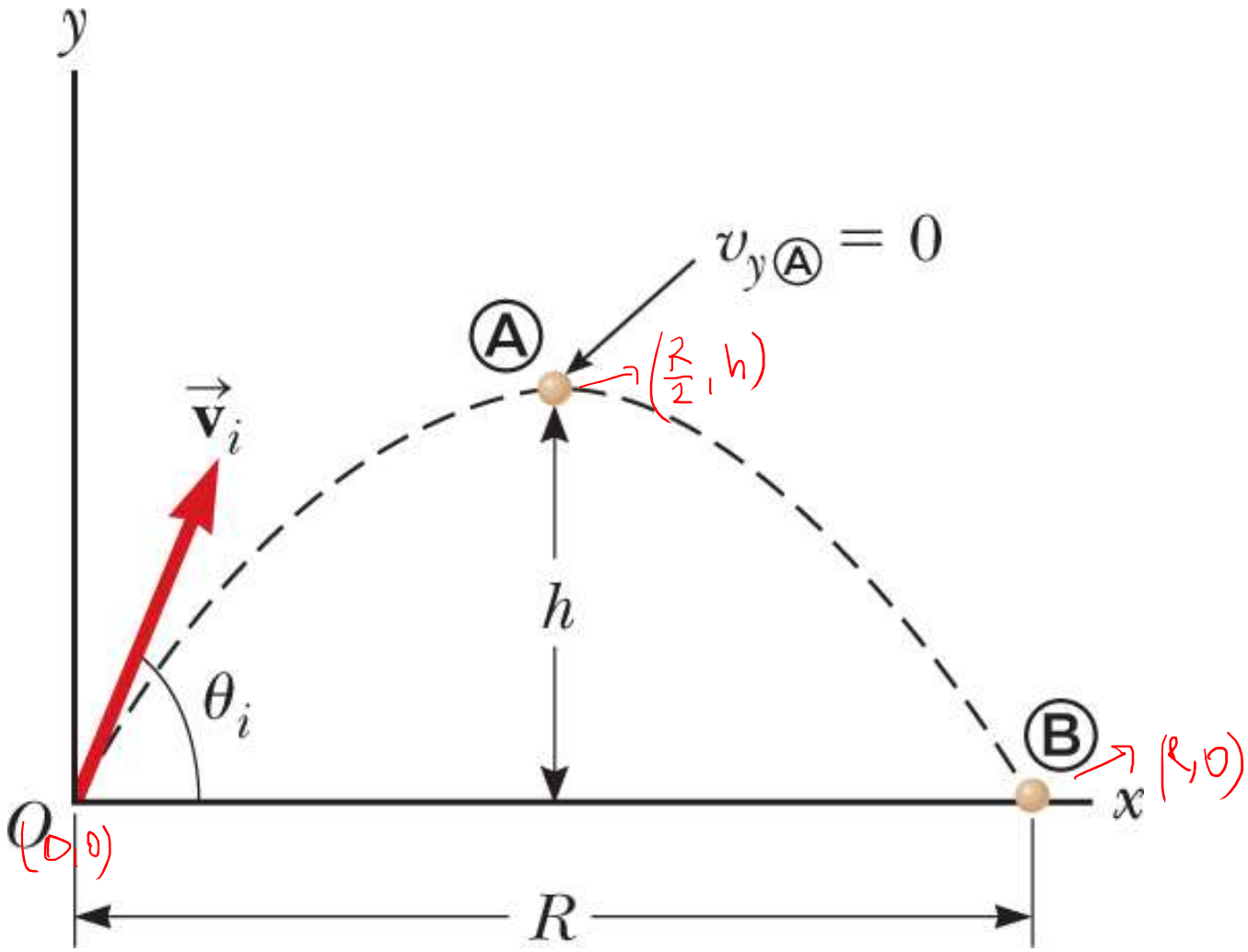
(a) nowhere

(b) the highest point

(c) the launch point

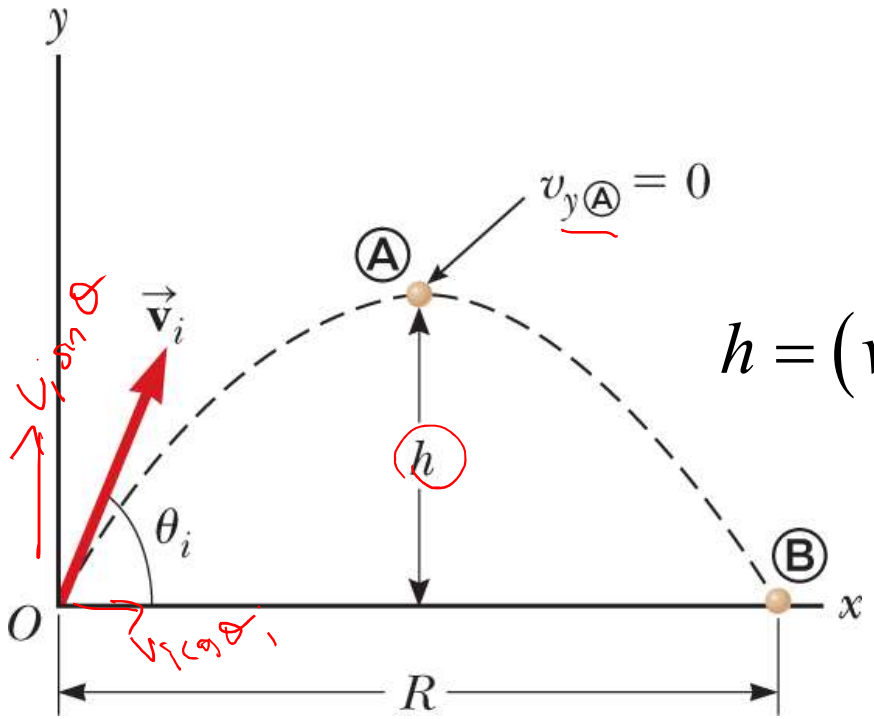


Horizontal Range and Maximum Height of a Projectile



Height of a Projectile

$$v_{yf} = v_{yi} - gt \Rightarrow 0 = v_i \sin\theta_i - gt_A \Rightarrow t_A = \frac{v_i \sin\theta_i}{g}$$



$$y_f = y_i + v_{yi}t - \frac{1}{2}gt^2 \Rightarrow$$

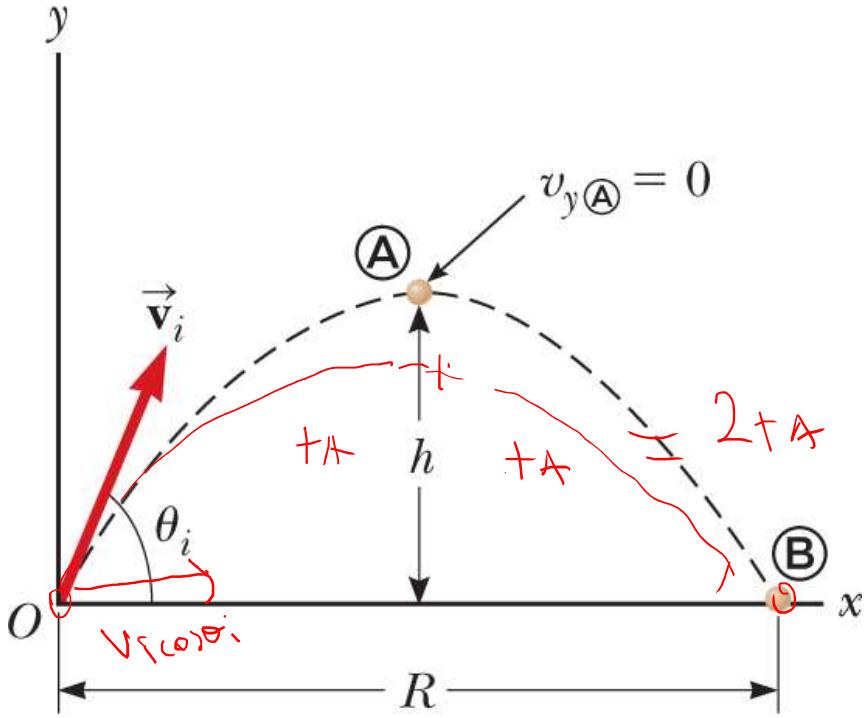
$$h = (v_i \sin\theta_i)\frac{v_i \sin\theta_i}{g} - \frac{1}{2}g\left(\frac{v_i \sin\theta_i}{g}\right)^2$$

$$h = \frac{v_i^2 \sin^2\theta_i}{2g}$$

Horizontal Range of a Projectile

$$x_f = x_i + v_{xi}t \Rightarrow R = v_{xi}t_B = \left(v_i \cos\theta\right) 2t_A$$

$$R = \left(v_i \cos\theta_i\right)\frac{2v_i \sin\theta_i}{g} = \frac{2v_i^{\,2} \sin\theta_i \cos\theta_i}{g}$$

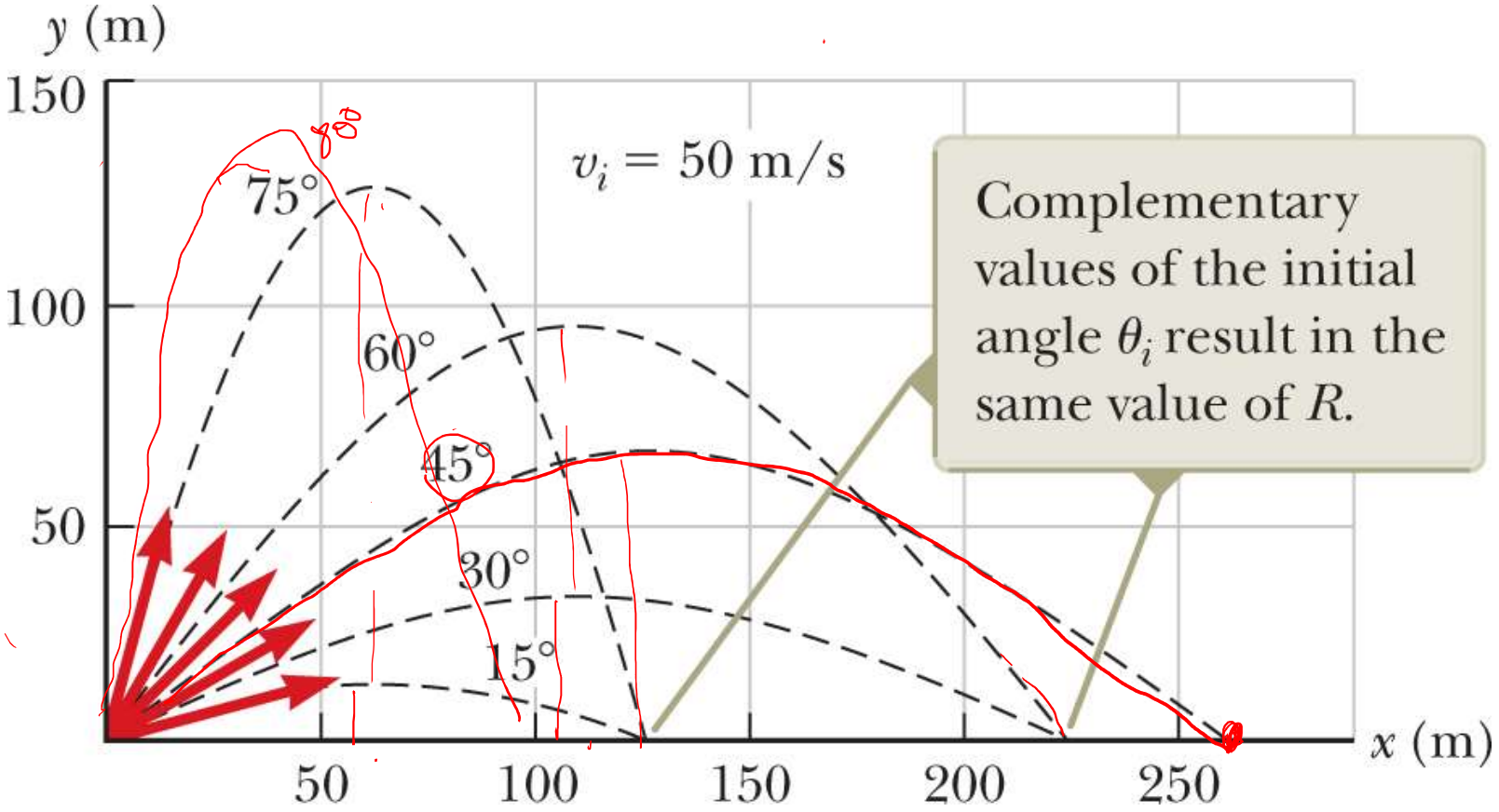


$$2\sin\theta_i \cos\theta_i = \sin 2\theta_i$$

$$R = \frac{v_i^{\,2} \sin 2\theta_i}{g}$$

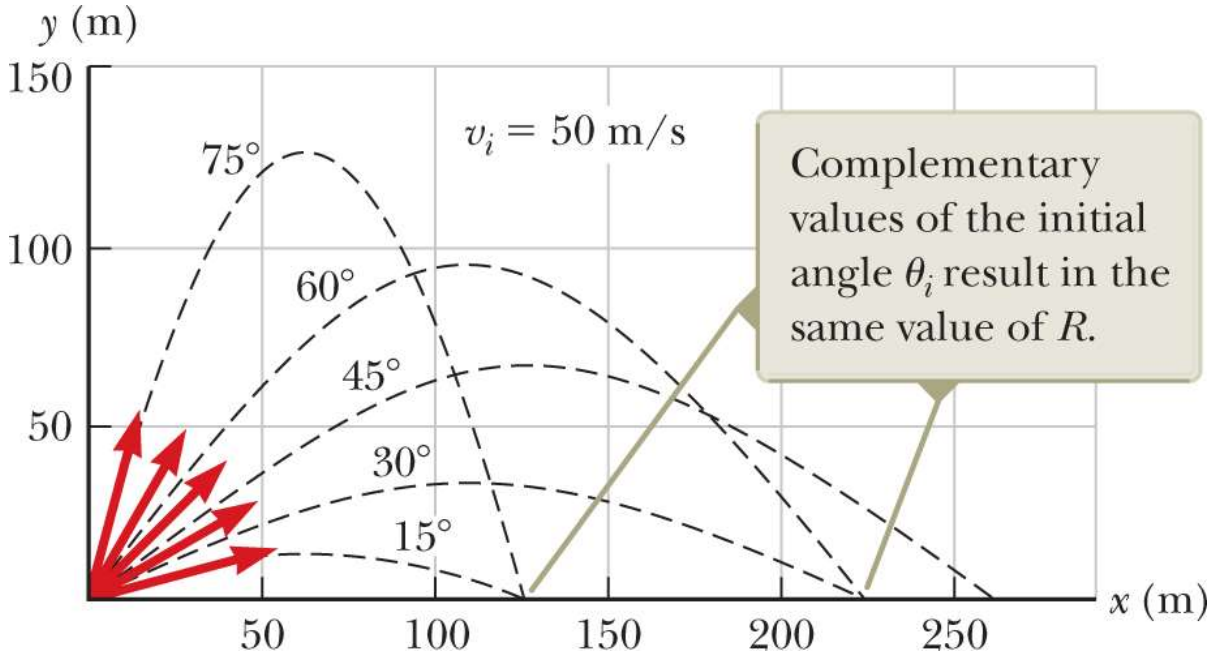
$$R_{\max} = \frac{v_i^{\,2}}{g}$$

Trajectories of a Projectile



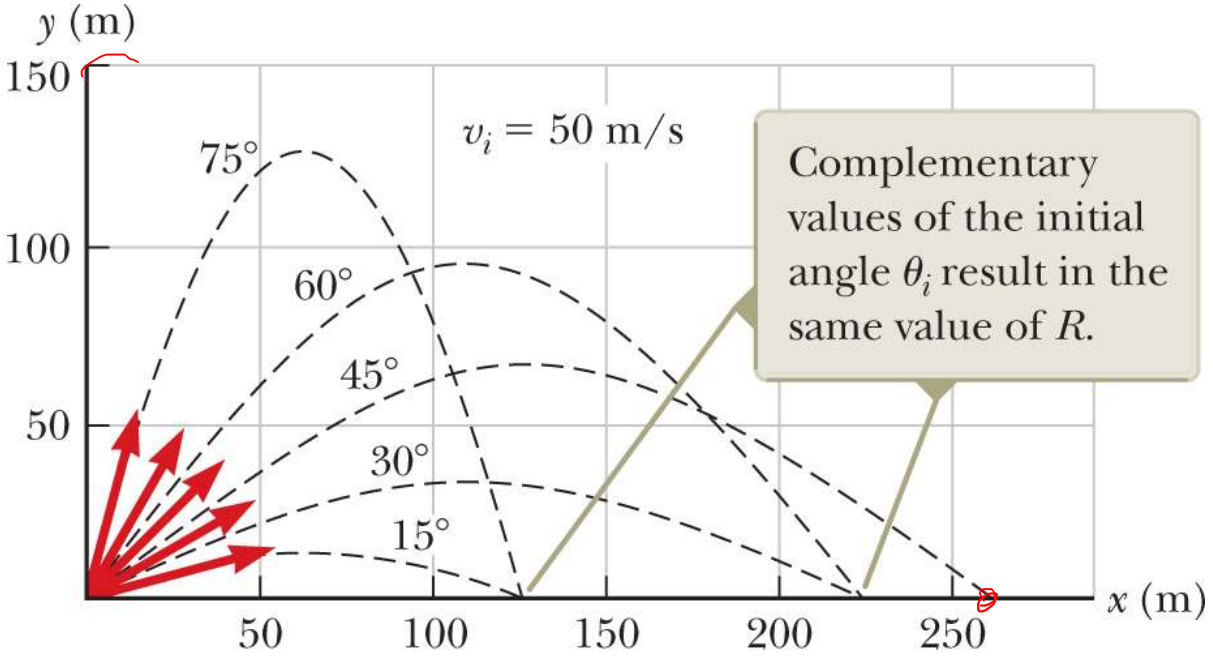
Quick Quiz 4.3

Rank the launch angles for the five paths in the figure with respect to time of flight from the shortest time of flight to the longest.



Quick Quiz 4.3

Rank the launch angles for the five paths in the figure with respect to time of flight from the shortest time of flight to the longest.



15°, 30°, 45°, 60°, 75°

Problem-Solving Strategy: Projectile Motion

1. Conceptualize

2. Categorize

$$h = \frac{v_i^2 \sin^2 \theta_i}{2g} \qquad R = \frac{v_i^2 \sin 2\theta_i}{g}$$

Problem-Solving Strategy: Projectile Motion

3. Analyze

$$x_f = x_i + v_x t$$

$$v_{yf} = v_{yi} - gt$$

$$v_{\text{y,avg}} = \frac{v_{yi} + v_{yf}}{2}$$

$$y_f = y_i + \frac{1}{2}\left(v_{yi} + v_{yf}\right)t$$

$$y_f = y_i + v_{yi}t - \frac{1}{2}gt^2$$

$$v_{yf}^{\ 2} = v_{yi}^{\ 2} - 2g\left(y_f - y_i\right)$$

4. Finalize

Example 4.2:
The Long Jump

A long jumper leaves the ground at an angle of 20.0° above the horizontal and at a speed of 11.0 m/s.
(A) How far does he jump in the horizontal direction?

$$R = \frac{v_i^2 \sin 2\theta_i}{g}$$

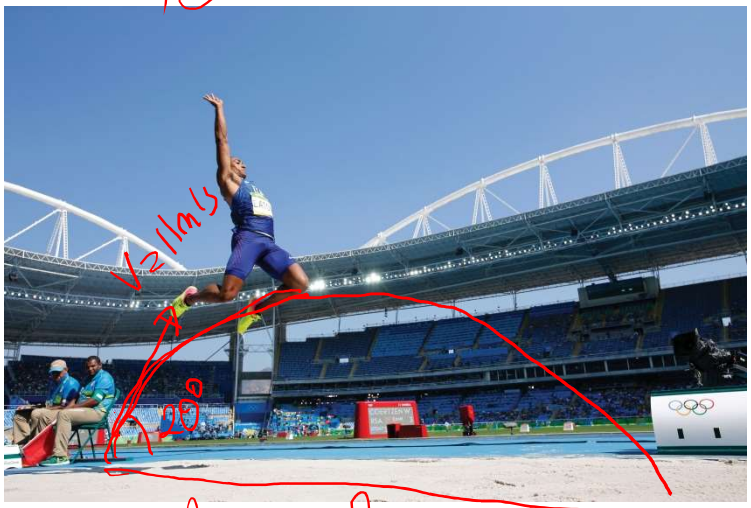
$$= \frac{(11.0 \text{ m/s})^2 \sin(2 \cdot 20.0°)}{9.80 \text{ m/s}^2}$$

$$= \boxed{7.94 \text{ m}}$$

$V_{fy} = V_{oy} - gt$

$0 = 11 \sin 20 - 9.8 \cdot t$

② $t = ?$



Example 4.2:
The Long Jump

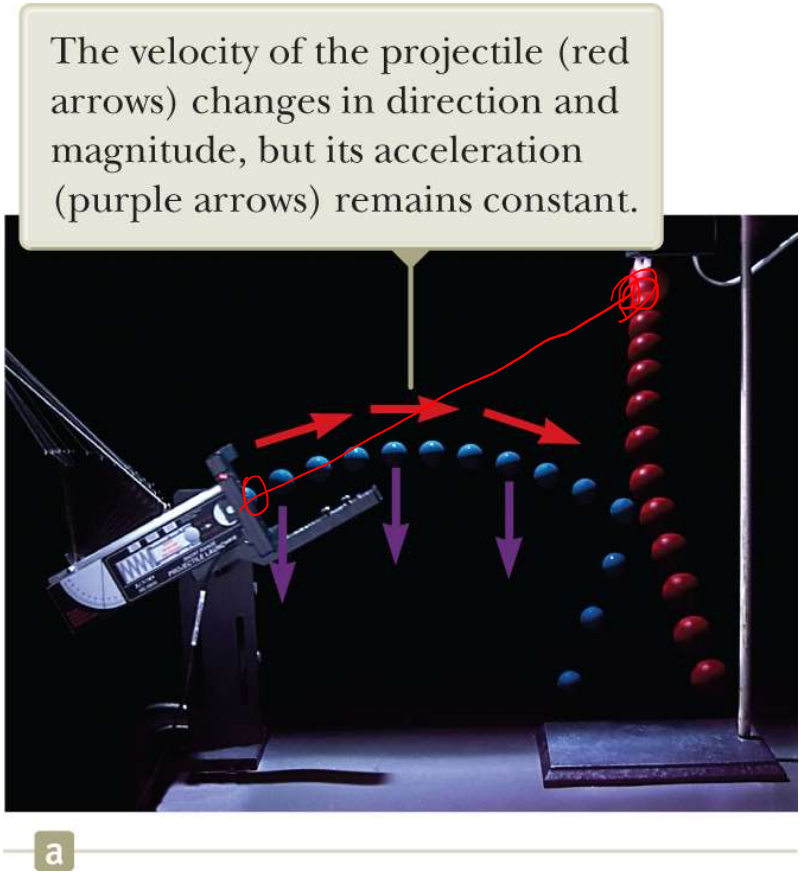
(B) What is the maximum height reached?

$$h = \frac{v_i^2 \sin^2 \theta_i}{2g} = \frac{\left(11.0 \text{ m/s}\right)^2 \left(\sin 20.0°\right)^2}{2\left(9.80 \text{ m/s}^2\right)}$$

$$= \boxed{0.722 \text{ m}}$$

Example 4.3:
A Bull's-Eye Every Time

In a popular lecture demonstration, a projectile is aimed directly at a target in such a way that the projectile leaves the gun at the same time the target is dropped from rest. Show that the projectile hits the falling target as shown in the figure.



Example 4.3: A Bull's-Eye Every Time

$$y_T = y_{iT} + (0)t - \frac{1}{2}gt^2 = x_T \tan\theta_i - \frac{1}{2}gt^2$$

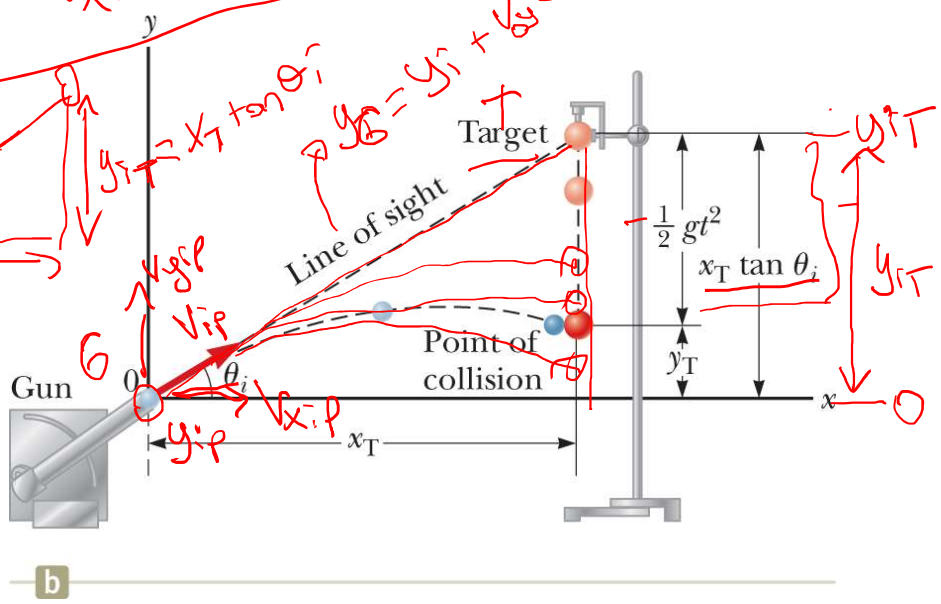
$$y_P = y_{iP} + v_{yiP}t - \frac{1}{2}gt^2 = 0 + (v_{iP}\sin\theta_i)t - \frac{1}{2}gt^2 = (v_{iP}\sin\theta_i)t - \frac{1}{2}gt^2$$

$$x_P = x_{iP} + v_{xiP}t = 0 + (v_{iP}\cos\theta_i)t = (v_{iP}\cos\theta_i)t$$

$$t = \frac{x_P}{v_{iP}\cos\theta_i}$$

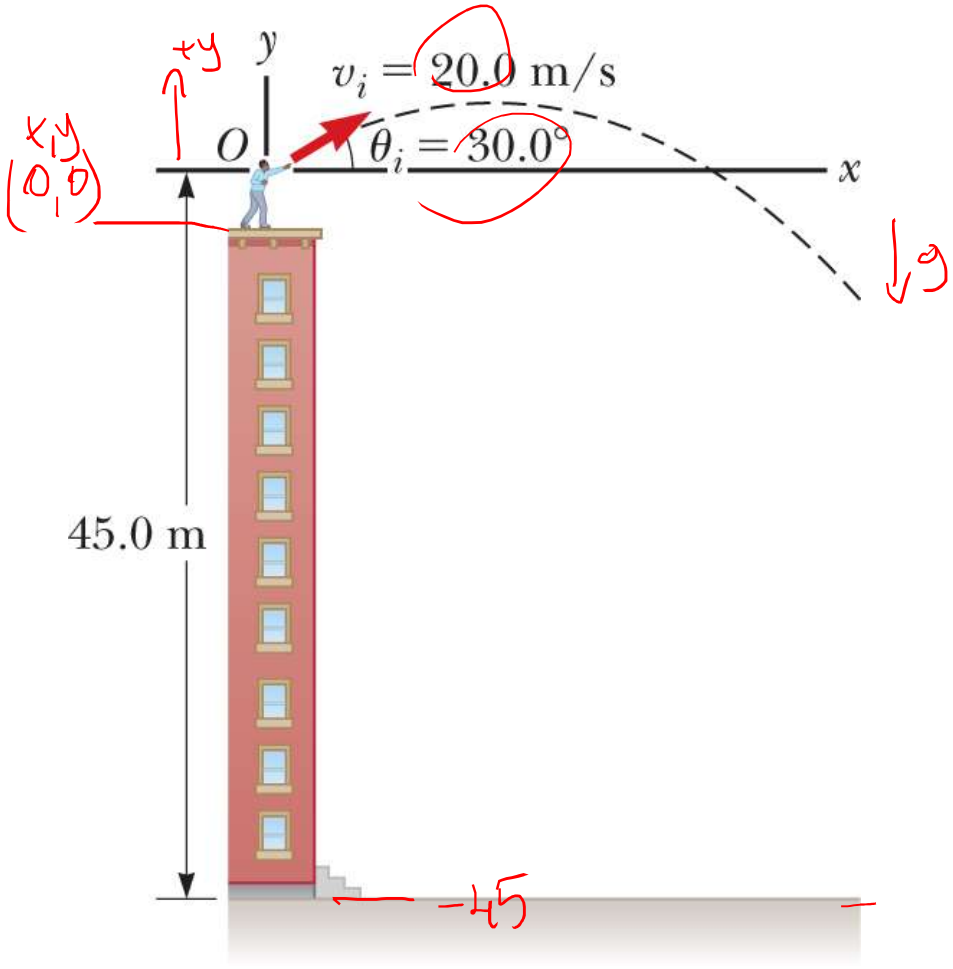
$$y_P = (v_{iP}\sin\theta_i)\left(\frac{x_P}{v_{iP}\cos\theta_i}\right) - \frac{1}{2}gt^2$$

$$y_P = x_P\tan\theta_i - \frac{1}{2}gt^2 = y_T$$



Example 4.4:
That's Quite an Arm

A stone is thrown from the top of a building upward at an angle of 30.0° to the horizontal with an initial speed of 20.0 m/s as shown in the figure. The height from which the stone is thrown is 45.0 m above the ground.



(A) How long does it take the stone to reach the ground?

Example 4.4:
That's Quite an Arm

$$x_i = y_i = 0,\ y_f = -45.0\ \text{m},\ a_y = -g,\ v_i = 20.0\ \text{m/s}$$

$$v_{xi} = v_i \cos\theta_i = (20.0\ \text{m/s})\cos 30° = 17.3\ \text{m/s}$$

$$v_{yi} = v_i \sin\theta_i = (20.0\ \text{m/s})\sin 30° = 10.0\ \text{m/s}$$

$$y_f = y_i + v_{yi}t - \frac{1}{2}gt^2$$

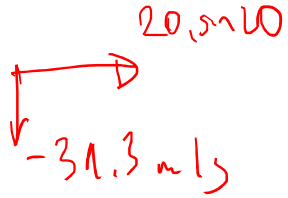
$$-45.0\ \text{m} = 0 + (10.0\ \text{m/s})t + \frac{1}{2}(-9.80\ \text{m/s}^2)t^2$$

$$t = \boxed{4.22\ \text{s}}$$

Example 4.4:
That's Quite an Arm

(B) What is the speed of the stone just before it strikes the ground?

$$v_{yf} = v_{yi} - gt$$



$$v_{yf} = 10.0 \text{ m/s} + \left(-9.80 \text{ m/s}^2\right)\left(4.22 \text{ s}\right) = -31.3 \text{ m/s}$$

$$v_f = \sqrt{v_{xf}^{\,2} + v_{yf}^{\,2}}$$

$$= \sqrt{\left(17.3 \text{ m/s}\right)^2 + \left(-31.3 \text{ m/s}\right)^2} = \boxed{35.8 \text{ m/s}}$$

Example 4.4:
That's Quite an Arm

What if a horizontal wind is blowing in the same direction as the stone is thrown and it causes the stone to have a horizontal acceleration component a_x = 0.500 m/s^2? Which part of this example, (A) or (B), will have a different answer?

Time does not change.

$v_{xf} = v_{xi} + a_x t$

$$a_x = 0.500 \text{ m/s}^2 \rightarrow v_{xf} = 19.4 \text{ m/s} \text{ and } v_f = 36.9 \text{ m/s}$$

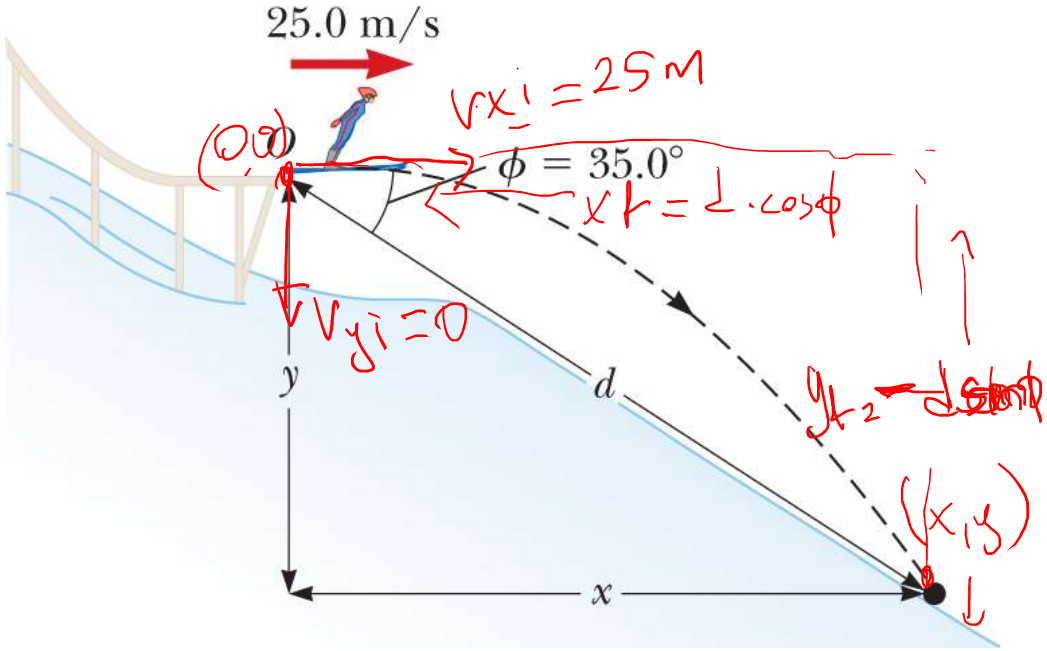
Final speed will be larger.

Example 4.5:
The End of the Ski Jump

A ski jumper leaves the ski track moving in the horizontal direction with a speed of 25.0 m/s as shown in the figure. The landing incline below her falls off with a slope of 35.0°. Where does she land on the incline?

$x_f = d\cos\phi$

$y_f = -d\cos\phi$



Example 4.5:
The End of the Ski Jump

$$x_f = d\cos\phi \quad \text{and} \quad y_f = -d\sin\phi$$

$$(1)\ x_f = v_{xi}t \rightarrow (2)\ d\cos\phi = v_{xi}t$$

$$(3)\ y_f = v_{yi}t - \frac{1}{2}gt^2 \rightarrow (4)\ -d\sin\phi = -\frac{1}{2}gt^2$$

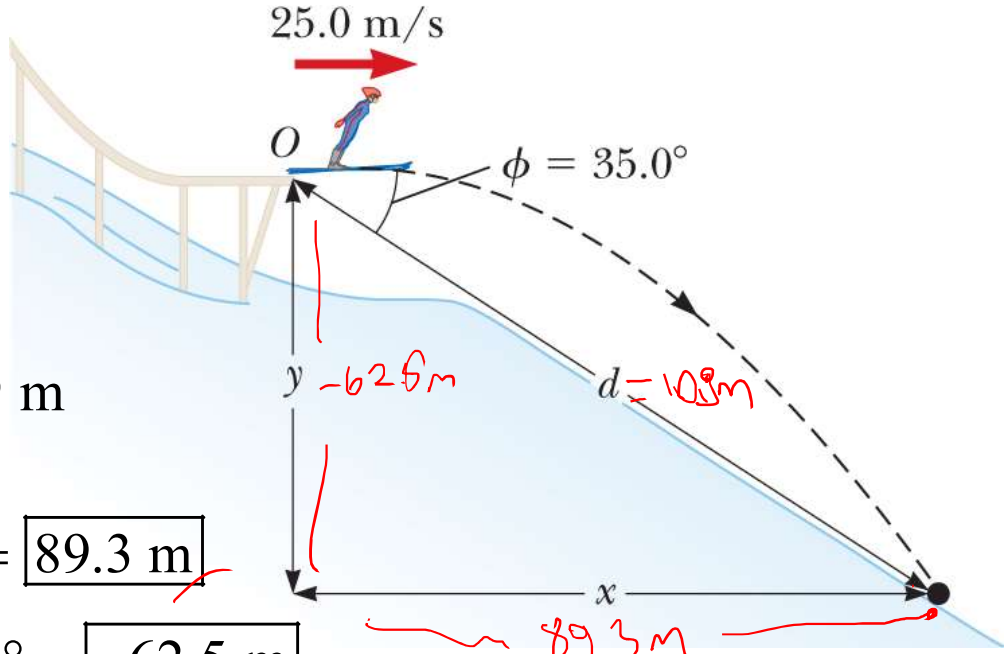
$$-d\sin\phi = -\frac{1}{2}g\left(\frac{d\cos\phi}{v_{xi}}\right)^2$$

$$d = \frac{2v_{xi}^{\,2}\sin\phi}{g\cos^2\phi}$$

$$= \frac{2(25.0\ \text{m/s})^2 \sin 35.0^\circ}{(9.80\ \text{m/s}^2)\cos^2 35.0^\circ} = 109\ \text{m}$$

$$x_f = d\cos\phi = (109\ \text{m})\cos 35.0^\circ = \boxed{89.3\ \text{m}}$$

$$y_f = -d\sin\phi = -(109\ \text{m})\sin 35.0^\circ = \boxed{-62.5\ \text{m}}$$



Example 4.5:
The End of the Ski Jump

Suppose everything in this example is the same except the ski jump is curved so that the jumper is projected upward at an angle from the end of the track. Is this design better in terms of maximizing the length of the jump?

$$x = (v_i \cos\theta)t \quad y = (v_i \sin\theta)t - \frac{1}{2}gt^2$$

$$(v_i \sin\theta)t - \frac{1}{2}gt^2 = (v_i \cos\theta)(\tan\phi)t$$

$$t = \frac{2v_i}{g}(\sin\theta - \cos\theta \tan\phi)$$

$$x = \frac{2v_i^2}{g}(\sin\theta\cos\theta - \cos^2\theta \tan\phi)$$

Example 4.5:
The End of the Ski Jump

$$x = \frac{2v_i^2}{g}\left(\sin\theta\cos\theta - \cos^2\theta\tan\phi\right)$$

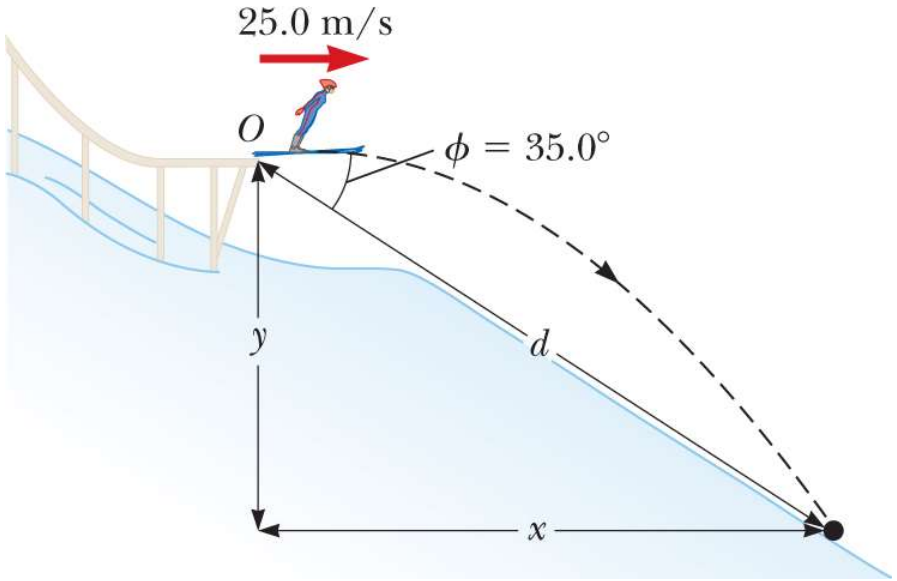
$$0 = \frac{d}{d\theta}\left(\frac{2v_i^2}{g}\left(\sin\theta\cos\theta - \cos^2\theta\tan\phi\right)\right)$$

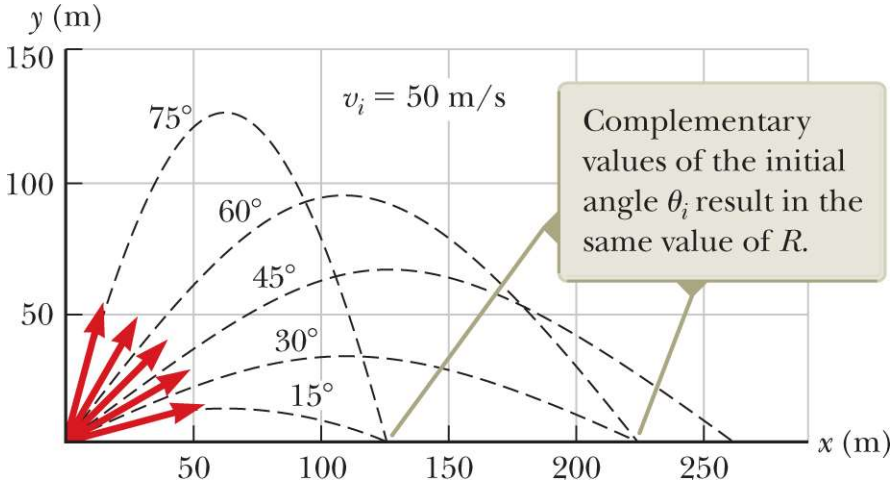
$$= \frac{2v_i^2}{g}\left(\cos^2\theta - \sin^2\theta - 2\cos\theta\sin\theta\tan\phi\right)$$

$$= \frac{2v_i^2}{g}\left(\cos(2\theta) - \sin(2\theta)\tan\phi\right)$$

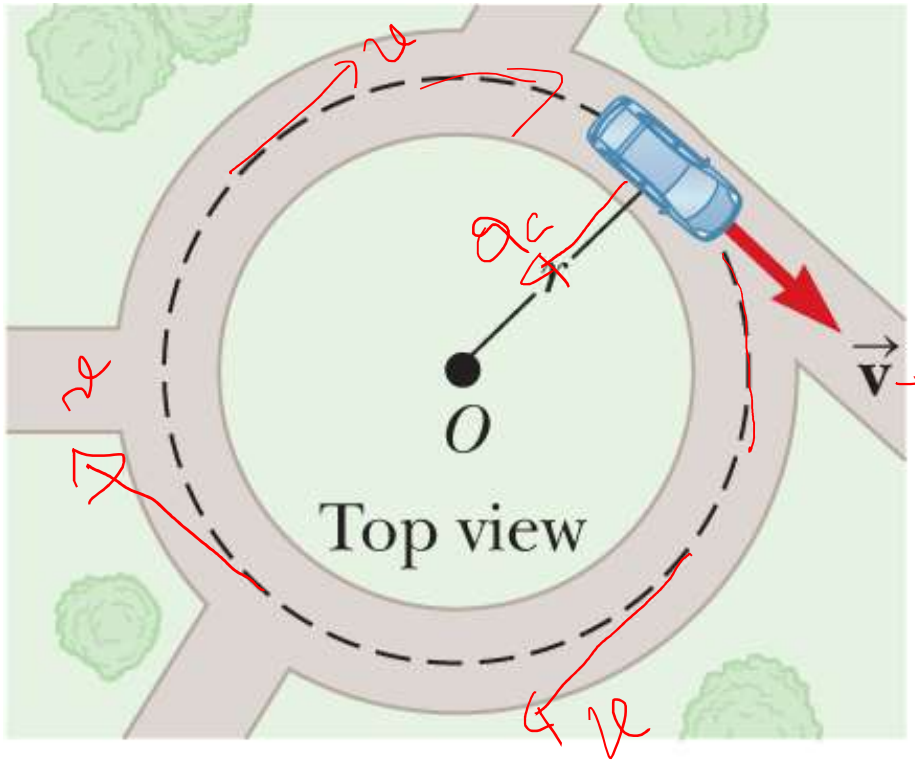
$$\tau = \frac{1}{2}\tan^{-1}\left(\cot\phi\right) = \frac{1}{2}\left(90^\circ - \phi\right) = 45^\circ - \frac{1}{2}\phi$$

Example 4.5:
The End of the Ski Jump





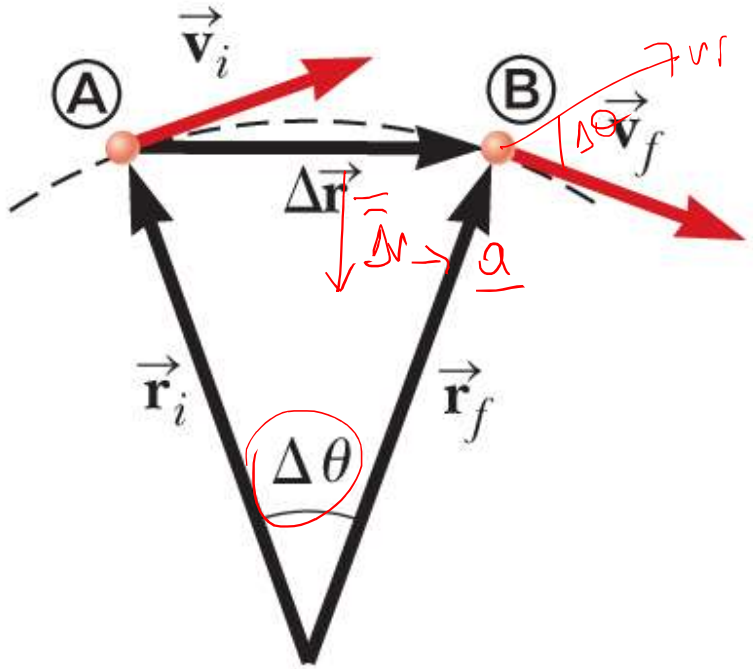
Analysis Model: Particle in Uniform Circular Motion

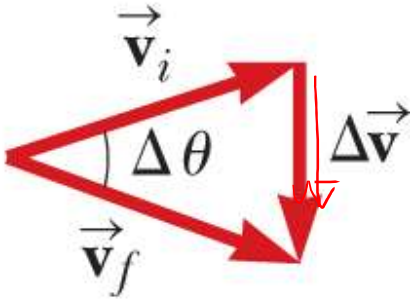


a

$$\vec{\mathbf{a}} = \frac{d\vec{\mathbf{v}}}{dt}$$

Analysis Model: Particle in Uniform Circular Motion





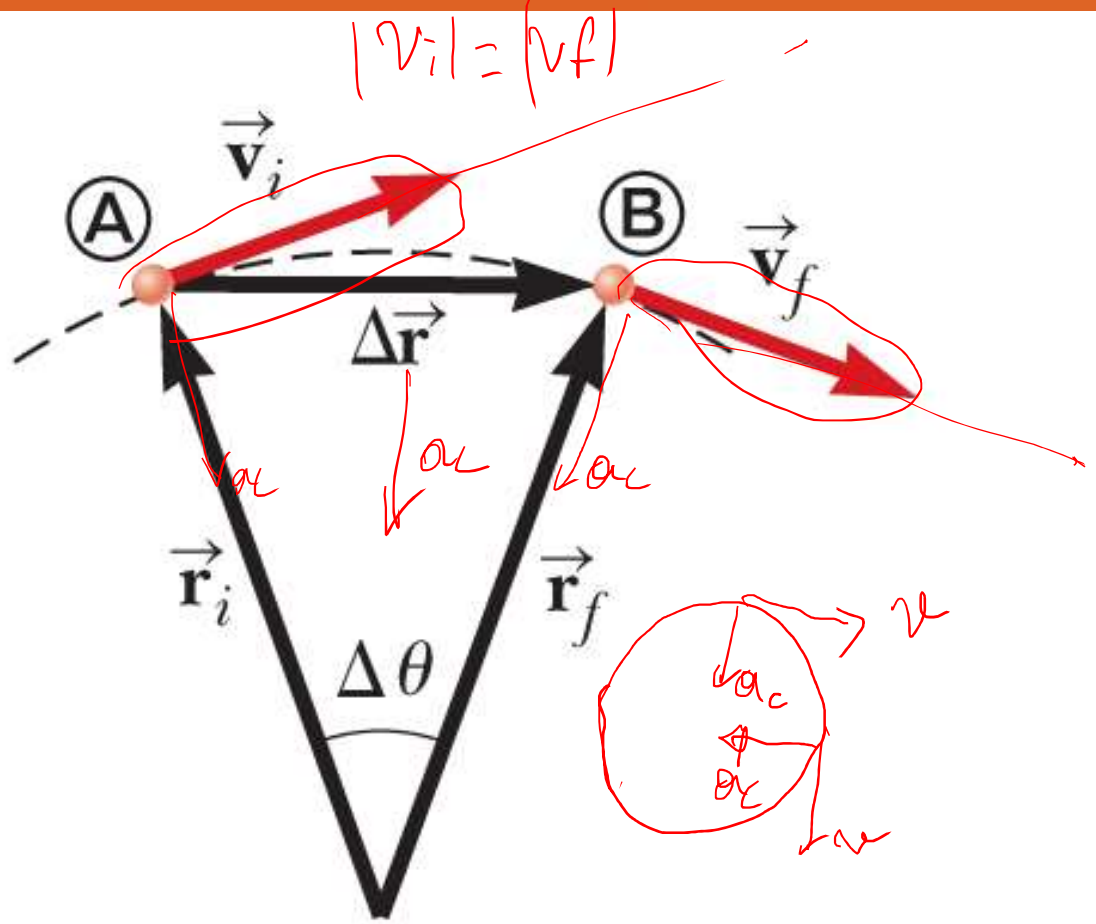
b

$$\frac{|\Delta\vec{\mathbf{v}}|}{v} = \frac{|\Delta\vec{\mathbf{r}}|}{r}$$

c

$$|\vec{\mathbf{a}}_{\text{avg}}| = \frac{|\Delta\vec{\mathbf{v}}|}{\Delta t} = \frac{v|\Delta\vec{\mathbf{r}}|}{r\Delta t}$$

Centripetal Acceleration



$$a_c = \frac{v^2}{r}$$

b

Particle in Uniform Circular Motion

$x = vt$

$2\pi r = v \cdot T \rightarrow$ period

T → time required for one full rotation.

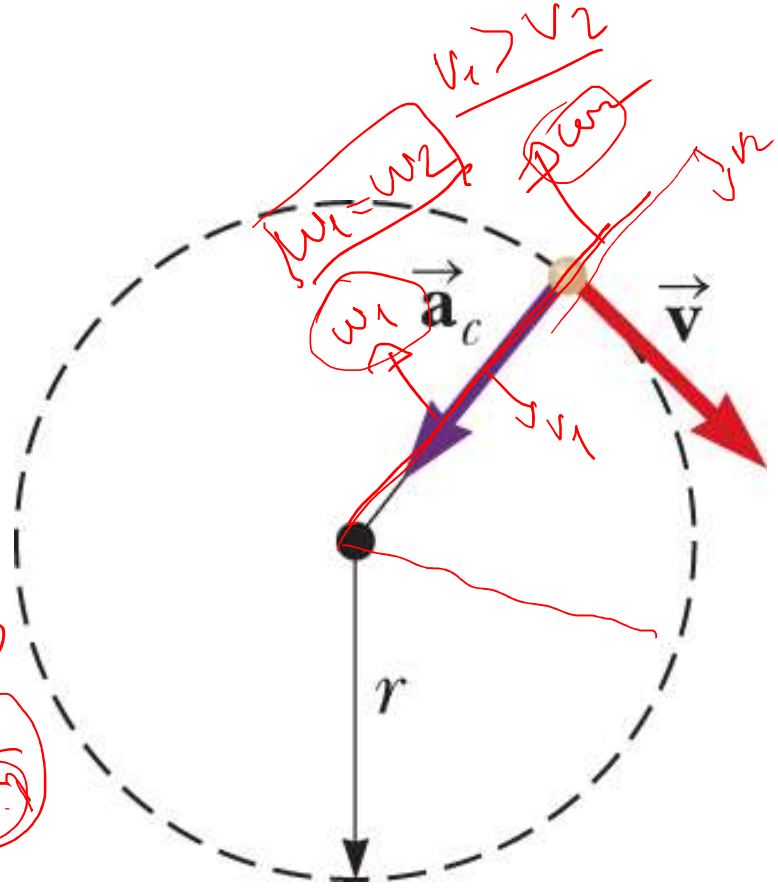
$$v = \frac{2\pi r}{T} \rightarrow \boxed{T = \frac{2\pi r}{v}}$$ → Period of circular motion.

$$\omega = \frac{2\pi}{T}$$ → angular velocity

$$\omega = 2\pi\left(\frac{v}{2\pi r}\right) = \frac{v}{r} \rightarrow \boxed{v = r\omega}$$

$$a_c = \frac{(r\omega)^2}{r} = r\omega^2$$

$a_c = \frac{v^2}{r}$



Quick Quiz 4.4 Part I

A particle moves in a circular path of radius r with speed v. It then increases its speed to $2v$ while traveling along the same circular path.

The centripetal acceleration of the particle has changed by what factor? Choose one:

$a_c = \frac{v^2}{r}$

(a) 0.25

(b) 0.5

(c) 2

(d) 4

(e) impossible to determine

Quick Quiz 4.4 Part I

A particle moves in a circular path of radius r with speed v. It then increases its speed to $2v$ while traveling along the same circular path.

The centripetal acceleration of the particle has changed by what factor? Choose one:

(a) 0.25

(b) 0.5

(c) 2

(d) 4

(e) impossible to determine

Quick Quiz 4.4 Part II

A particle moves in a circular path of radius r with speed v. It then increases its speed to $2v$ while traveling along the same circular path.

By what factor has the period of the particle changed?

Choose one:

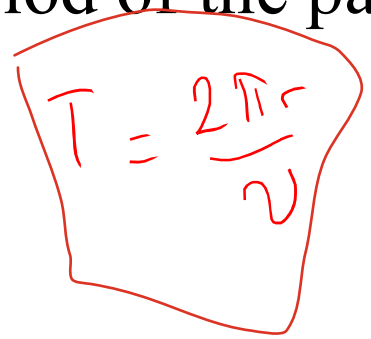
(a) 0.25

(b) 0.5

(c) 2

(d) 4

(e) impossible to determine



Quick Quiz 4.4 Part II

A particle moves in a circular path of radius r with speed v. It then increases its speed to $2v$ while traveling along the same circular path.

By what factor has the period of the particle changed?

Choose one:

(a) 0.25

(b) 0.5

(c) 2

(d) 4

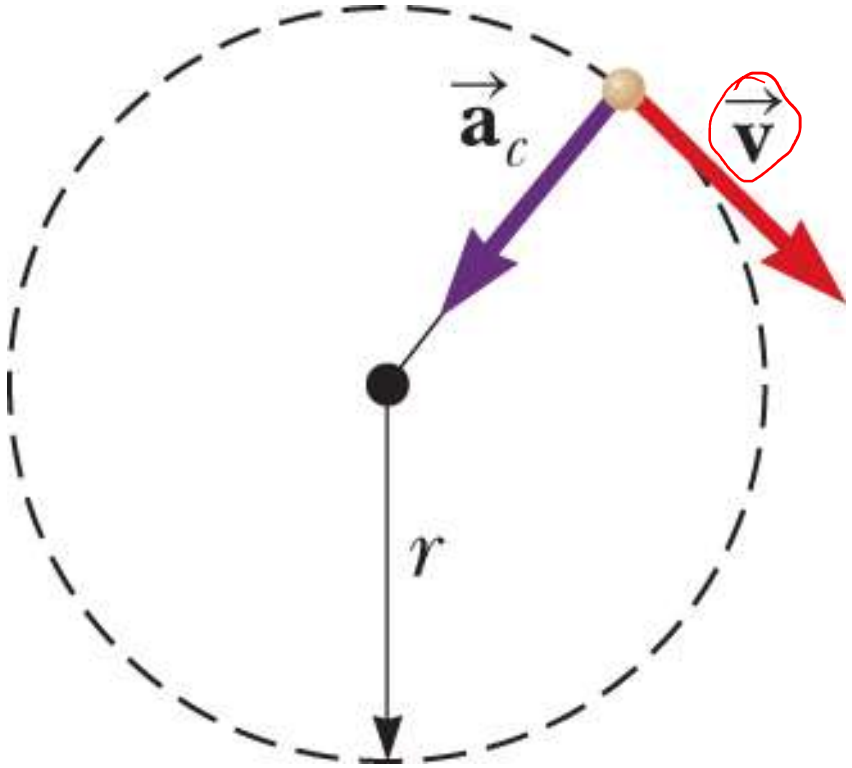
(e) impossible to determine

Analysis Model:
Particle in Uniform Circular Motion

$$a_c = \frac{v^2}{r}$$

$$T = \frac{2\pi r}{v}$$

$$\omega = \frac{2\pi}{T}$$



Example 4.6: The Centripetal Acceleration of the Earth

(A) What is the centripetal acceleration of the Earth as it moves in its orbit around the Sun?

$g = 10\ \text{m/s}^2$

$$a_c = \frac{v^2}{r} = \frac{\left(\frac{2\pi r}{T}\right)^2}{r} = \frac{4\pi^2 r}{T^2}$$

$$a_c = \frac{4\pi^2 \left(1.496\times10^{11}\ \text{m}\right)}{\left(1\ \text{yr}\right)^2}\left(\frac{1\ \text{yr}}{3.156\times10^{7}\ \text{s}}\right)^2 = \boxed{5.93\times10^{-3}\ \text{m/s}^2}$$

Example 4.6: The Centripetal Acceleration of the Earth

(B) What is the angular speed of the Earth in its orbit around the Sun?

$\omega = \frac{2\pi}{T}$

$$\omega = \frac{2\pi}{1\ \text{yr}}\left(\frac{1\ \text{yr}}{3.156\times10^{7}\ \text{s}}\right) = \boxed{1.99\times10^{-7}\ \text{s}^{-1}}$$

Tangential and Radial Acceleration

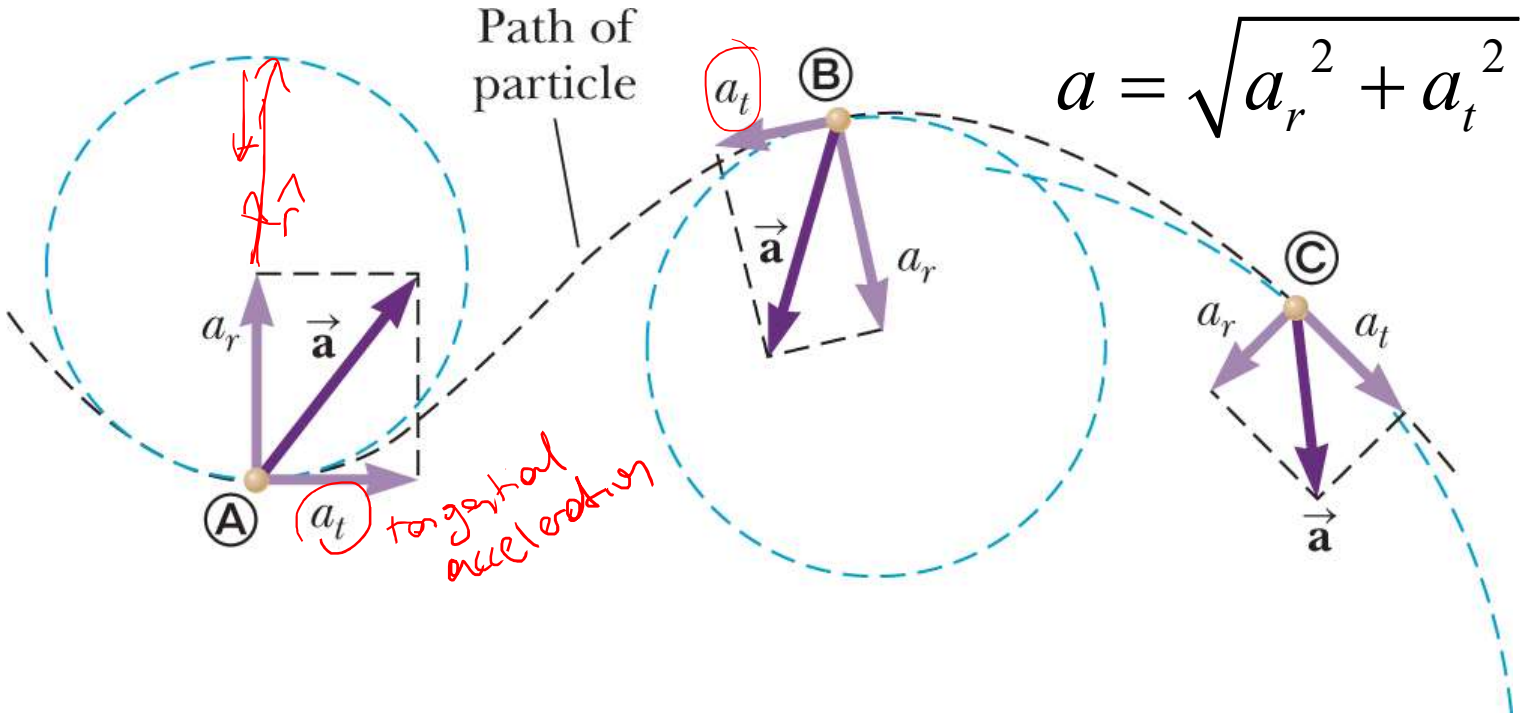
$a_c = a_r$ → Radial acceleration

Total acceleration

$$\vec{\mathbf{a}} = \vec{\mathbf{a}}_r + \vec{\mathbf{a}}_t$$

For non-uniform circular motion.

$$a_t = \left| \frac{dv}{dt} \right| \qquad a_r = -a_c = -\frac{v^2}{r}$$



$$a = \sqrt{a_r^{\,2} + a_t^{\,2}}$$

Quick Quiz 4.5 Part I

A particle moves along a path, and its speed increases with time.

In which of the following cases are its acceleration and velocity vectors parallel?

(a) when the path is circular

(b) when the path is straight

(c) when the path is a parabola

(d) never

Quick Quiz 4.5 Part I

A particle moves along a path, and its speed increases with time.

In which of the following cases are its acceleration and velocity vectors parallel?

(a) when the path is circular

(b) when the path is straight

(c) when the path is a parabola

(d) never



Quick Quiz 4.5 Part II

A particle moves along a path, and its speed increases with time.

In which case are its acceleration and velocity vectors perpendicular everywhere along the path?

(a) when the path is circular

(b) when the path is straight

(c) when the path is a parabola

(d) never

Quick Quiz 4.5 Part II

A particle moves along a path, and its speed increases with time.

In which case are its acceleration and velocity vectors perpendicular everywhere along the path?

(a) when the path is circular

(b) when the path is straight

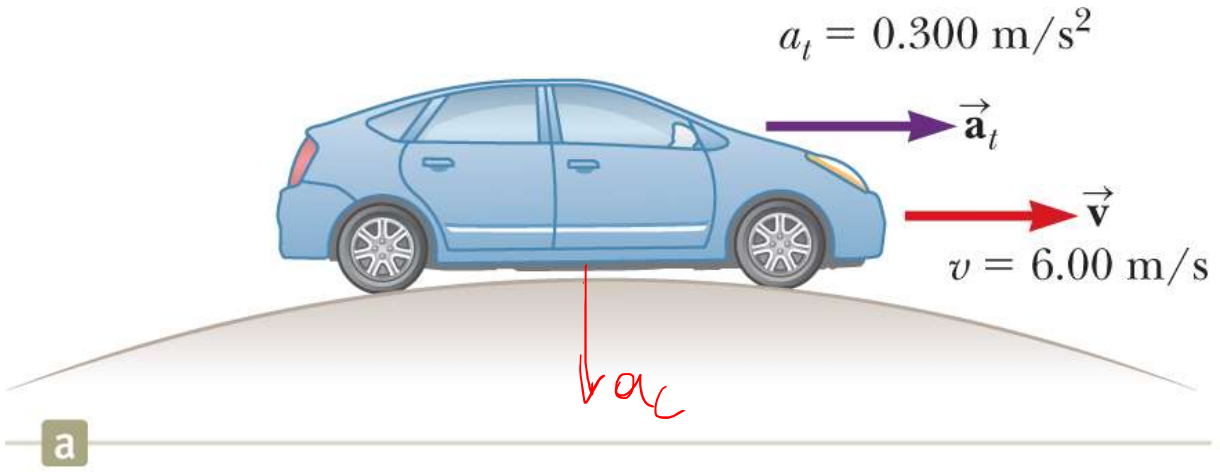
(c) when the path is a parabola

(d) never



Example 4.7: Over the Rise

A car leaves a stop sign and exhibits a constant acceleration of 0.300 m/s^2 parallel to the roadway. The car passes over a rise in the roadway such that the top of the rise is shaped like an arc of a circle of radius 500 m. At the moment the car is at the top of the rise, its velocity vector is horizontal and has a magnitude of 6.00 m/s. What are the magnitude and direction of the total acceleration vector for the car at this instant?

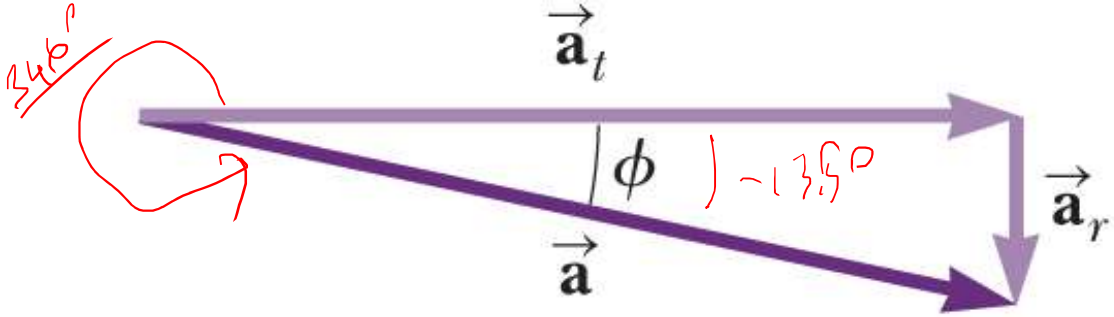


Example 4.7: Over the Rise

$$a_r = -\frac{v^2}{r} = -\frac{(6.00\ \text{m/s})^2}{500\ \text{m}} = -0.0720\ \text{m/s}^2$$

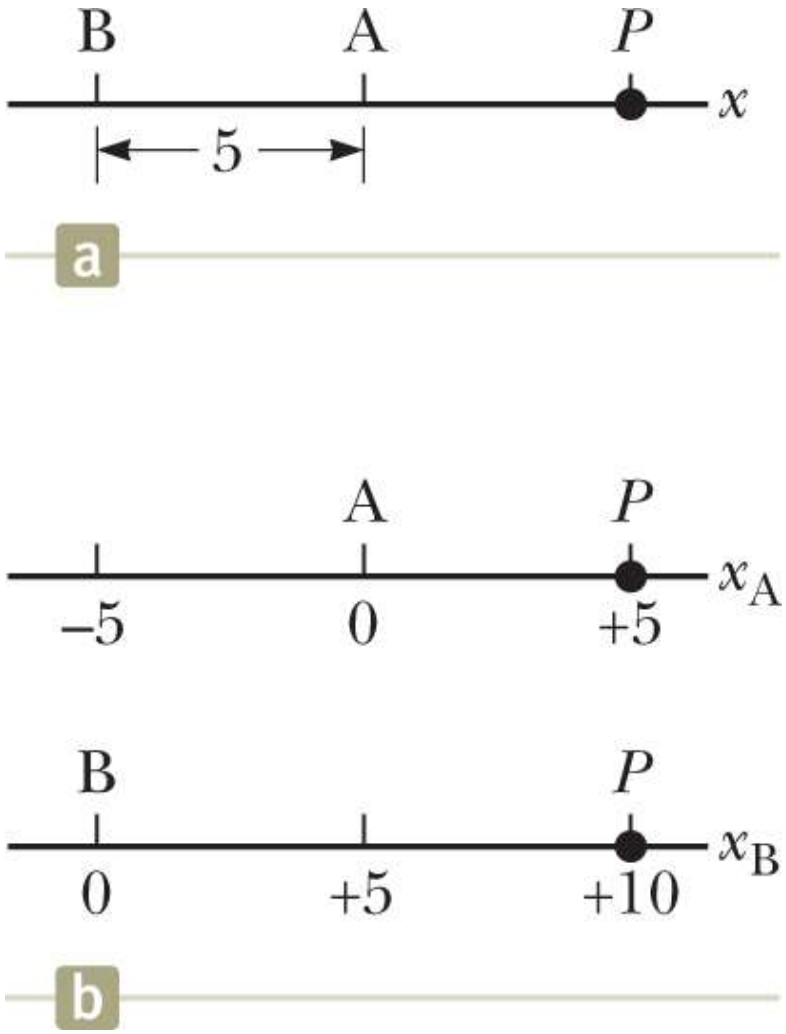
$$a = \sqrt{a_r^{\,2} + a_t^{\,2}}$$

$$= \sqrt{(-0.0720\ \text{m/s}^2)^2 + (0.300\ \text{m/s}^2)^2} = \boxed{0.309\ \text{m/s}^2}$$

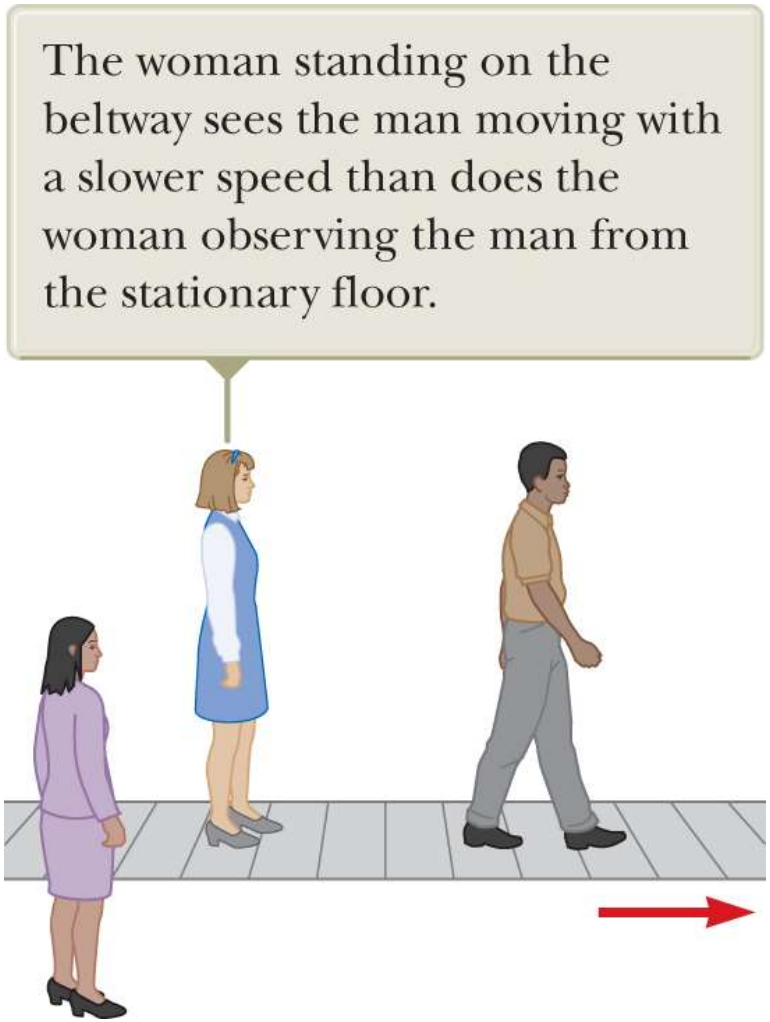


$$\phi = \tan^{-1}\frac{a_r}{a_t} = \tan^{-1}\left(\frac{-0.0720\ \text{m/s}^2}{0.300\ \text{m/s}^2}\right) = \boxed{-13.5^\circ}$$

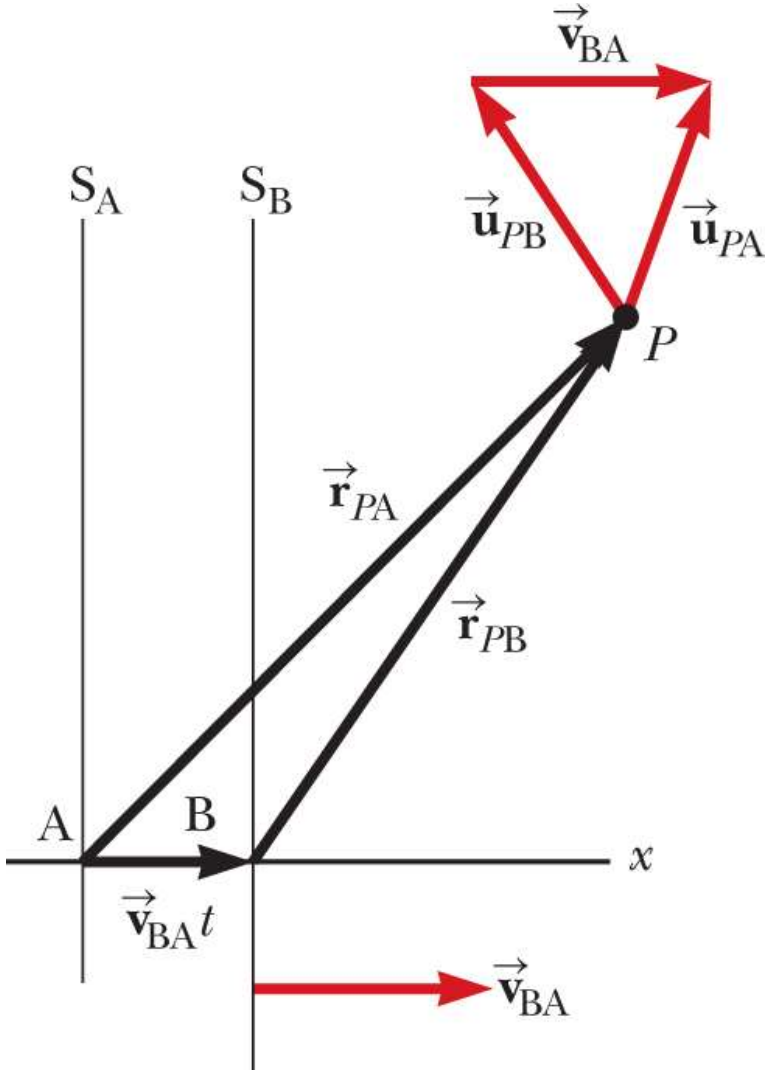
Relative Velocity and Relative Acceleration



Relative Velocity and Relative Acceleration



Relative Velocity



$$\vec{\mathbf{r}}_{PA} = \vec{\mathbf{r}}_{PB} + \vec{\mathbf{v}}_{BA}t$$

$$\frac{d\vec{\mathbf{r}}_{PA}}{dt} = \frac{d\vec{\mathbf{r}}_{PB}}{dt} + \vec{\mathbf{v}}_{BA}$$

$$\vec{\mathbf{u}}_{PA} = \vec{\mathbf{u}}_{PB} + \vec{\mathbf{v}}_{BA}$$

Relative Acceleration

$$\vec{\mathbf{u}}_{PA} = \vec{\mathbf{u}}_{PB} + \vec{\mathbf{v}}_{BA} \rightarrow \qquad \frac{d\vec{\mathbf{u}}_{PA}}{dt} = \frac{d\vec{\mathbf{u}}_{PB}}{dt} + \frac{d\vec{\mathbf{v}}_{BA}}{dt}$$

$$\frac{d\vec{\mathbf{v}}_{BA}}{dt} = 0 \rightarrow \frac{d\vec{\mathbf{u}}_{PA}}{dt} = \frac{d\vec{\mathbf{u}}_{PB}}{dt}$$

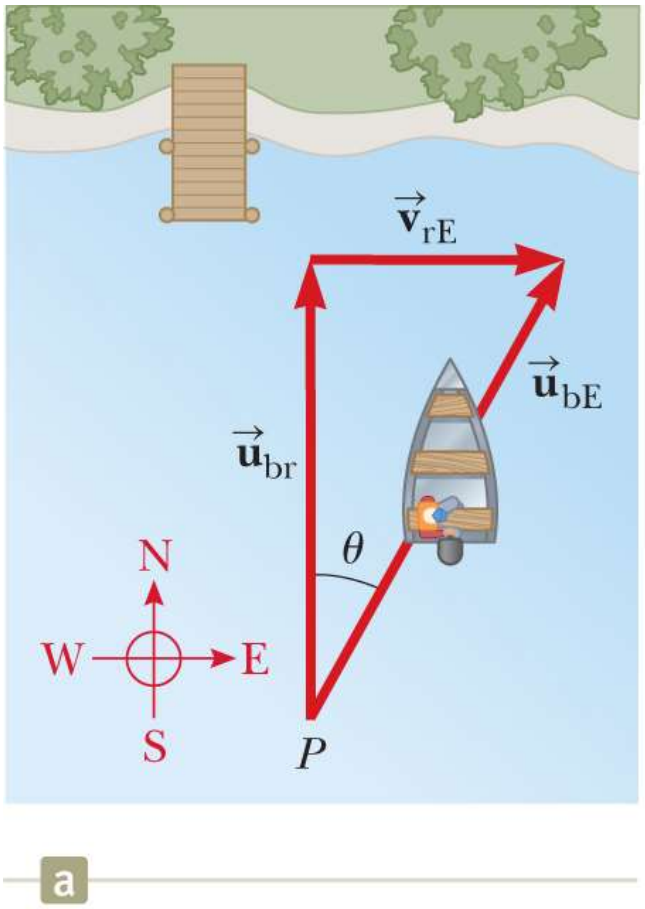
$$\vec{\mathbf{a}}_{PA} = \frac{d\vec{\mathbf{u}}_{PA}}{dt} \text{ and } \vec{\mathbf{a}}_{PB} = \frac{d\vec{\mathbf{u}}_{PB}}{dt}$$

$$\vec{\mathbf{a}}_{PA} = \vec{\mathbf{a}}_{PB}$$

Example 4.8:
A Boat Crossing a River

A boat crossing a wide river moves with a speed of 10.0 km/h relative to the water. The water in the river has a uniform speed of 5.00 km/h due east relative to the Earth.

(A) If the boat heads due north, determine the velocity of the boat relative to an observer standing on either bank.



Example 4.8:
A Boat Crossing a River

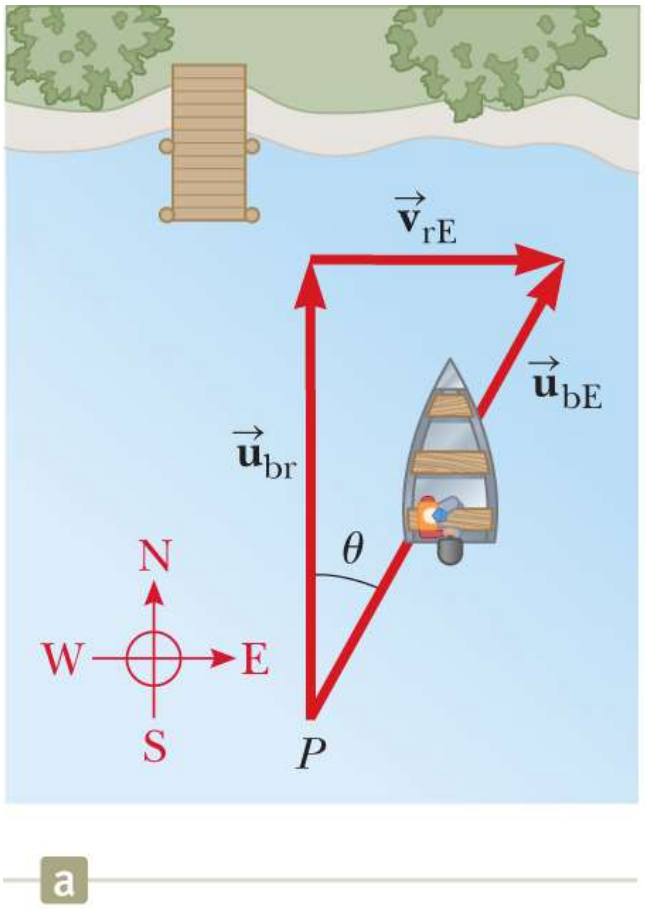
$$u_{bE} = \sqrt{u_{br}^{\,2} + u_{rE}^{\,2}}$$

$$= \sqrt{(10.0\ \text{km/h})^2 + (5.00\ \text{km/h})^2}$$

$$= \boxed{11.2\ \text{km/h}}$$

$$\theta = \tan^{-1}\left(\frac{v_{rE}}{u_{br}}\right)$$

$$= \tan^{-1}\left(\frac{5.00}{10.0}\right) = \boxed{26.6^\circ}$$



a

Example 4.8:
A Boat Crossing a River

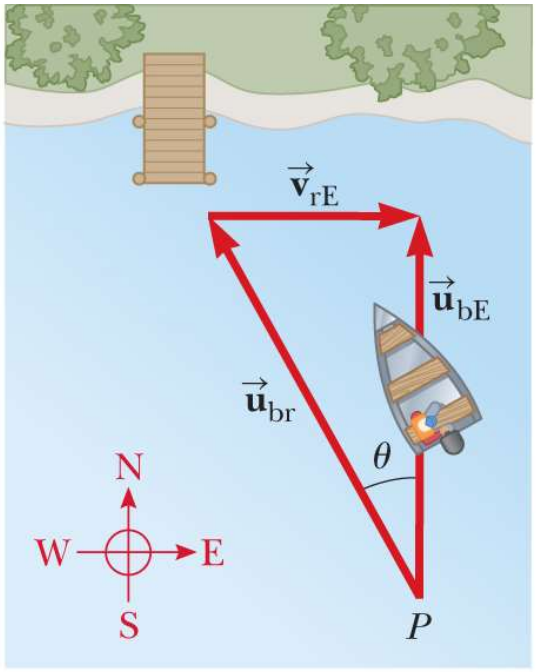
(B) If the boat travels with the same speed of 10.0 km/h relative to the river and is to travel due north as shown in the figure, what should its heading be?

$$u_{\text{bE}} = \sqrt{u_{\text{br}}{}^2 - v_{\text{rE}}{}^2}$$

$$= \sqrt{(10.0\ \text{km/h})^2 - (5.00\ \text{km/h})^2}$$

$$= \boxed{8.66\ \text{km/h}}$$

$$\theta = \tan^{-1}\left(\frac{v_{\text{rE}}}{u_{\text{br}}}\right) = \tan^{-1}\left(\frac{5.00}{8.66}\right) = \boxed{30.0°}$$



b

Example 4.8:
A Boat Crossing a River

Imagine that the two boats in parts (A) and (B) are racing across the river. Which boat arrives at the opposite bank first?

Assessing to Learn

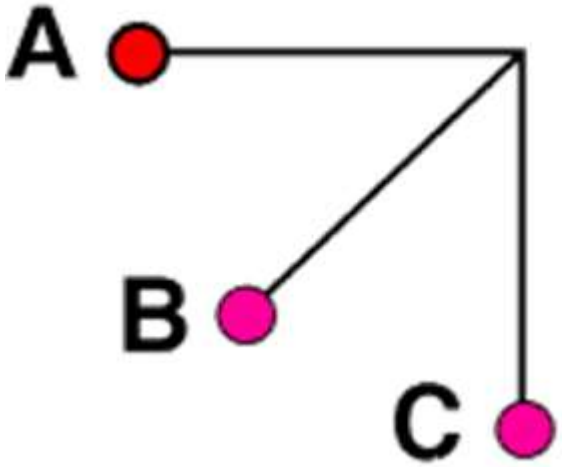
Consider the following situations:

- a car slowing down at a stop sign
- a ball being swung in a circle at constant speed
- a vibrating string
- the Moon orbiting the Earth
- a skydiver falling at terminal speed
- an astronaut in an orbiting space station
- a ball rolling down a hill
- a person driving down a straight section of highway at constant speed with her foot on the accelerator
- a molecule in the floor of this room

In how many of the situations is the object accelerating?

Assessing to Learn

A pendulum is released from rest at position A and swings toward the vertical under the influence of gravity as depicted below. When at position B, which direction most nearly corresponds to the direction of the acceleration?





Assessing to Learn

Consider the following situations:

- a car slowing down at a stop sign
- a ball being swung in a circle at constant speed
- a vibrating string
- the Moon orbiting the Earth
- a skydiver falling at terminal speed
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